SKAI

SILENT KILLER AI

The group guide: what every part of the app does, how it fits a South African clinic, what it cannot do yet, and how it could enter the health system, starting in Tshwane.

Read this before the presentation. Each feature comes with a screenshot, a plain explanation and a realistic clinic scenario. Nothing here assumes medical or programming knowledge.

Version as presented, October 2026 · all patients, staff and clinics shown are fictional demo data

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PART 1

Understanding SKAI

The problem, the idea, and the words you need

1. SKAI in one page

Heart disease and stroke are called "silent killers" because the damage builds up for years, mostly from high blood pressure, high cholesterol, diabetes and smoking, while the person feels completely fine. Many South Africans with high blood pressure do not know they have it, and many who do know are not controlling it. The first sign is often a heart attack or a stroke.

Public clinics are busy. Nurses see the person in front of them with today's complaint: a cough, a sprained ankle, a repeat prescription. The person whose blood pressure has been creeping up for years can easily walk out with nothing done, and never be called back.

SKAI (Silent Killer AI) is a web app that helps clinics find the people at highest risk of heart disease early, and makes sure they are booked, referred, reminded and followed up. It has two sides:

Side	Who it is for	What it does
Clinician dashboard	Nurses, doctors, facility managers, district managers	Waiting room triage (SATS), heart-risk assessment with explanations, priority bookings, referrals with letters, SMS reminders, lab results (FHIR), outcome tracking and monitoring for accuracy and bias.
SKAI Health (patient app)	Patients registered at the clinic	A personal heart-health tracker: risk from trusted clinic results only, a validated second score, tips, medication planner, daily check-ins, appointments, referrals, privacy controls. In all 11 official languages.

What SKAI is NOT

- It is **not a diagnosis**. It estimates risk. A clinician always decides.
- It does **not replace triage**. The South African Triage Scale decides how urgent someone is today; SKAI only works inside it.
- It is **not yet approved for real patients**. It is a working prototype that would need local testing and SAHPRA registration first (see Part 5 and Part 6).

The idea in one sentence

"Triage decides how urgent you are today; SKAI makes sure the people with the most dangerous hidden problem are not missed, are booked, and stay in care."

2. The demo world used in this guide

To make the app realistic without using anyone's real data, SKAI starts with a small fictional health district. In a real Tshwane deployment these would be, for example, a community clinic in Mamelodi or Soshanguve and the district hospital its patients are referred to.

Facility (fictional)	Type	Role in the story
Kopanong Community Clinic	Primary health care clinic	Most scenarios happen here: waiting room, bookings, referrals
Ubuntu Community Health Centre	Community health centre	A second clinic; has a patient whose document needs checking
Lesedi District Hospital	District hospital	Receives referrals from the clinics
Demo District	District office	The district manager sees all three facilities

The people

Name	Account	Role and facility
Sr Thandi Mokoena	nurse.thandi	Professional nurse, Kopanong
Dr Sfiso Kekana, Dr Mpho Mutengwe	sfiso.kekana, mpho.mutengwe	Doctors, Kopanong
Sr Lindiwe Zulu	nurse.lindiwe	Professional nurse, Ubuntu
Dr Kamogelo, Dr Neontle	kamogelo, neontle	Doctors, Ubuntu
Dr Samuel Olajide, Dr Tshiamo Bokaba	samuel.olajide, tshiamo.bokaba	Doctors, Lesedi District Hospital
Ms Refilwe Dube	manager.district	District manager (all facilities)

All staff use the password **SkaiDemo2026!**. Demo patients who have the app: **DEMO-101** (John Doe) and **DEMO-102** (Nomvula Khumalo), password **SkaiPatient2026!**. Other demo patients: Thabo Ndlovu (DEMO-001, critical risk), Sipho Mokoena (DEMO-002), Sarah Venter (DEMO-003), Pieter Botha (DEMO-004), Lerato Dlamini (DEMO-005).

3. A patient's journey through SKAI

Every feature in this guide is one step of the same journey. Keep this picture in mind:

1. Arrive Nurse records vitals; SATS colour → **2. Assess** Heart-risk estimate and what drives it → **3. Act** Book, refer, give app access → **4. Remind** SMS for visits and medicine → **5. Track** Patient app: results, meds, check-ins → **6. Learn** Record outcomes; monitor accuracy and bias

Scenario: Thabo's year with SKAI

- **Monday 08:10**: Thabo (58) comes to Kopanong clinic for a repeat prescription. Sr Thandi triages him: green, safe to wait. SKAI shows a red heart icon next to his name because his last assessment was critical, so he is seen first among the greens.
- **09:00**: Dr Kekana opens his record: 96.8% estimated risk, driven by abnormal heart tests and chest pain on exertion. He refers Thabo to Lesedi District Hospital as urgent and prints the referral letter.
- **Same day**: Thabo has opted in to SMS, so he gets a message telling him where to go and by when.
- **Three days later**: the hospital has not confirmed he arrived. The referral turns red as *overdue* in the clinic's queue, so the clinic phones him instead of losing him.
- **Months later**: an angiogram confirms coronary artery disease. Dr Kekana records the outcome, and SKAI's monitoring counts this as a correct prediction.

4. Key ideas in plain words

Risk percentage and risk tiers

SKAI's models give a number between 0% and 100%: how strongly this person's measurements resemble people in the training data who had heart disease. To make it easy to act on, the number is put into a tier:

Tier	SKAI estimate	What it means in practice	Next check-up (patient app)
Low	below 30%	Routine care and healthy-living advice	in 12 months
Moderate	30% to under 60%	Watch closely, treat risk factors	in 6 months

Tier	SKAI estimate	What it means in practice	Next check-up (patient app)
High	60% to under 85%	Should be booked within 14 days	in 3 months
Critical	85% and above	Should be booked within 7 days; consider referral	in 1 month

The two risk numbers (important for the presentation)

Patient results show **two** numbers, and they can be very different. For John Doe, SKAI says about 44%, while the Framingham score says about 9%. That is expected:

	SKAI's own estimate	Framingham 2008 score
Where it comes from	SKAI's machine-learning model, trained on data	A published formula from a large US study, typed into the app
What it is good for	**Ranking** patients: who is riskier than whom	A realistic **10-year chance** of heart attack or stroke
Why it can look high	The lifestyle model learned from SAHeart, a study that deliberately included many men who already had heart disease (about 1 in 3), so it over-estimates	It was built on ordinary people followed for years, so its percentages are realistic (for Americans)
Limit	Not yet tested on Tshwane patients	Built on Americans; not validated in South Africans

> **How to say it**
>
> "SKAI's number is for ranking patients, not a true percentage. We show the Framingham score next to it as the well-known reference. Once clinics record real outcomes, SKAI's monitoring shows whether its numbers need adjusting for South African patients."

SATS, TEWS and colours

The **South African Triage Scale (SATS)** is the tool South African emergency units and many clinics use to decide who must be seen first. A nurse measures breathing rate, heart rate, blood pressure, temperature, alertness (AVPU) and mobility. These give a **TEWS** score (Triage Early Warning Score). Certain danger signs, called **discriminators** (for example chest pain or not breathing), push the colour up immediately.

Colour	TEWS	See within	Example
Red	7 or more, or a red sign	Immediately	Not breathing, seizure now, very low blood sugar
Orange	5 to 6, or an orange sign	10 minutes	Chest pain, shortness of breath, stroke signs
Yellow	3 to 4, or a yellow sign	60 minutes	Moderate pain, vomiting, abdominal pain
Green	0 to 2	4 hours	Sprained ankle, repeat prescription

Trusted vitals

In a medical app, the numbers behind a risk score must come from a real measurement. That is why **patients can never type their own blood pressure or cholesterol**. They upload the document their clinic, pharmacy clinic or lab gave them, and SKAI reads it. Staff enter vitals during consultations, and lab systems can send results directly (FHIR).

Explainable AI (the “what is driving this” chart)

SKAI does not just give a number. For each patient it shows which measurements pushed the risk up (red bars) and which pulled it down (green or teal bars). This uses a method called **SHAP**. A clinician can then check that the reasoning makes medical sense instead of trusting a black box.

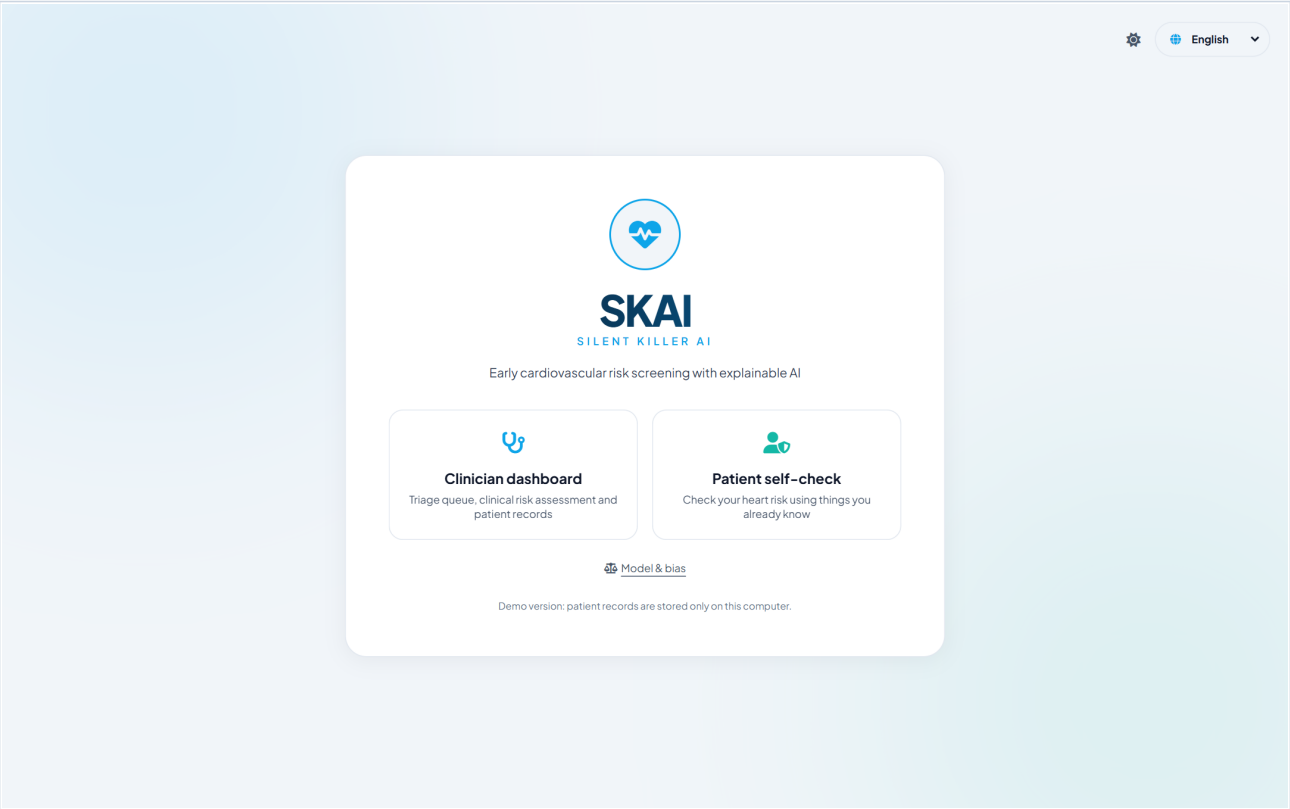
PART 2

Getting in

Home screen, sign-in, roles, security, themes and languages

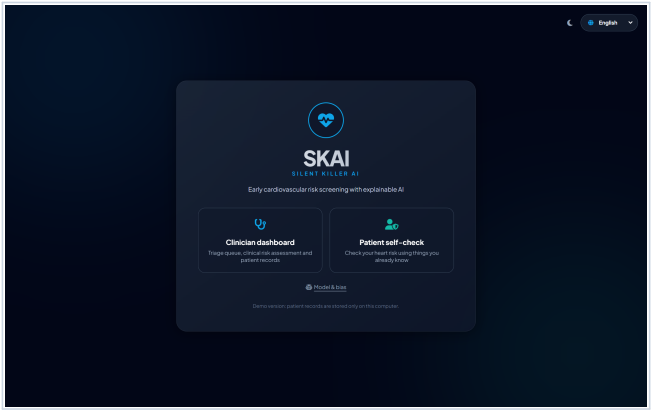
5. Home screen

Who uses it: Everyone | **Where in the app:** The first page at http://127.0.0.1:8000

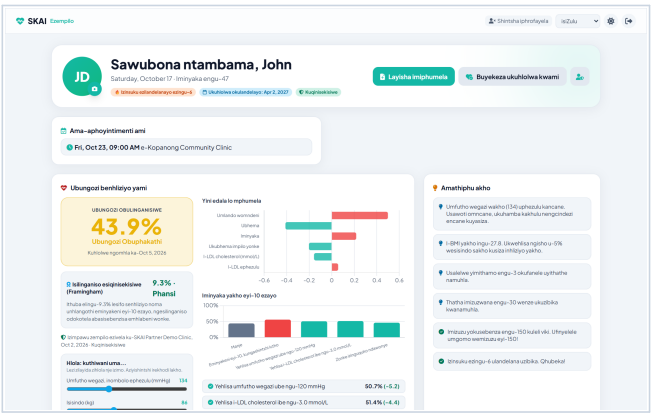


The home screen: choose the clinician dashboard or the patient app (SKAI Health).

- **Clinician dashboard**: for staff. Requires signing in.
- **Patient self-check**: opens SKAI Health, the patient app. Requires the patient's own sign-in.
- **Model & bias**: a public page showing how accurate the models are and whether they treat groups fairly (section 18).
- **Language** (top right) switches the whole app between the 11 official languages; the **moon/sun** button switches between dark and light theme.



Dark theme (default on devices set to dark mode).



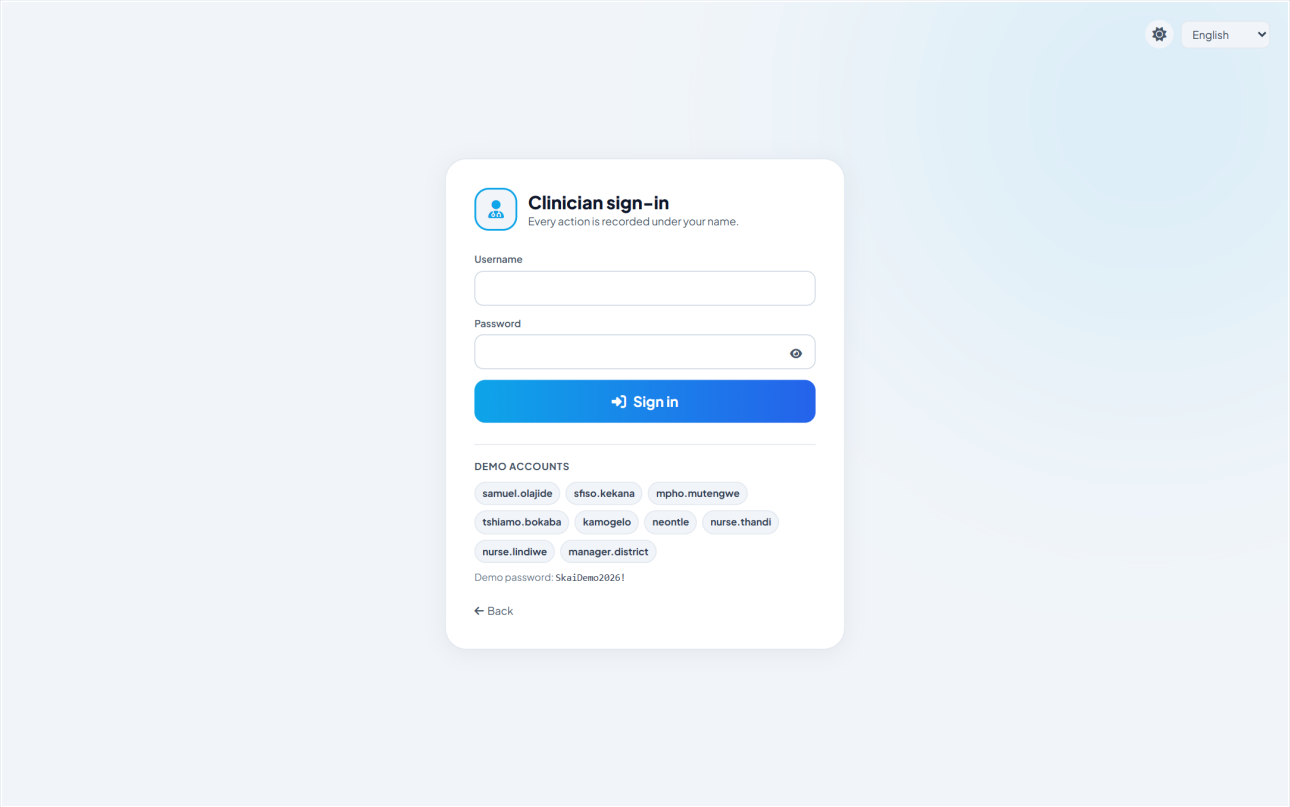
The patient app in isiZulu. Every patient-facing word is translated.

Limits to be honest about

The translations were written for this project and still need checking by native speakers before real patients rely on them. Staff-only screens added recently (waiting room, bookings, referrals, monitoring) are in English.

6. Staff sign-in, roles and security

Who uses it: All staff **Where in the app:** Home → Clinician dashboard



Clinician sign-in. The demo account buttons fill in the username; the password is SkaiDemo2026!.

Every staff member has their own account. That matters medically and legally: every action (opening a record, saving an assessment, approving a document, referring a patient) is recorded under that person's name.

Roles: who can do what

Action	Nurse	Doctor	Facility manager	District manager
Triage arrivals in the waiting room	Yes	Yes	No	No
Run risk assessments	Yes	Yes	No	No
Approve or reject patient documents	No	Yes	No	No
Book appointments	Yes	Yes	Yes	No
Refer patients	Yes	Yes	No	No
Give a patient an app access code	Yes	Yes	Yes	No
Record confirmed outcomes	No	Yes	No	No
See the Monitoring dashboard	No	Yes	Yes	Yes (all facilities)
Which patients they can see	Own facility + patients referred to it	Same	Same	Everyone in the district

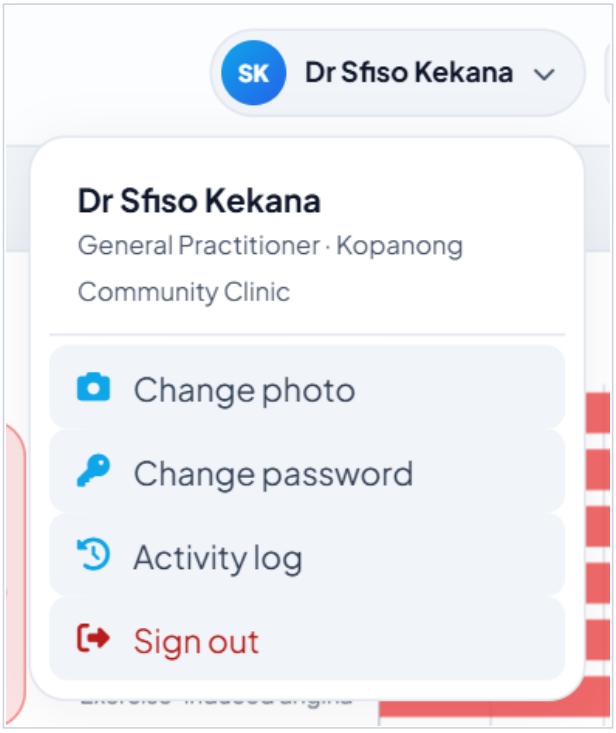
Scenario: Why the facility rule matters

Dr Samuel Olajide works at Lesedi District Hospital. He cannot open the records of Kopanong clinic patients, until Kopanong refers one of them to his hospital. From that moment he can see that patient's record, and when the referral is closed his access ends again. This is the "need to know" principle that the POPIA privacy law expects.

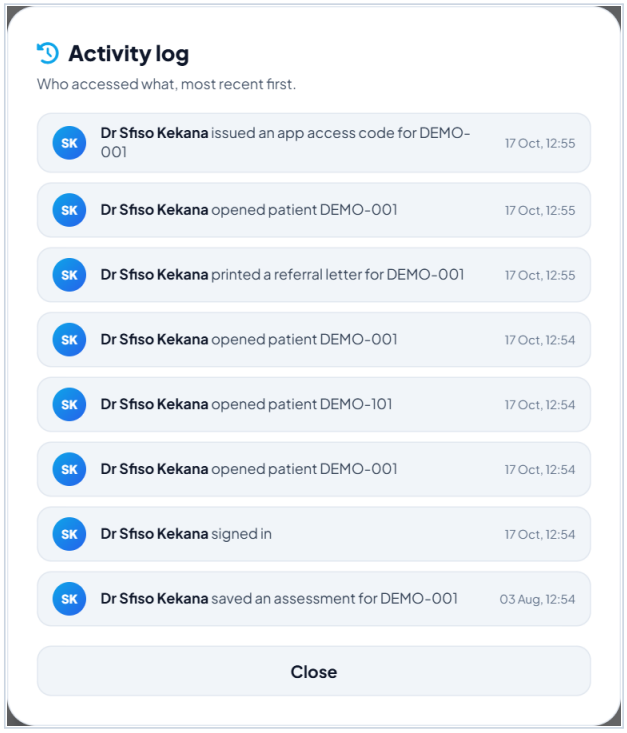
Security built in

- Passwords are never stored, only a scrambled "hash" (PBKDF2-SHA256), so even someone who copies the database cannot read them.
- Five wrong passwords lock the account for 5 minutes, which stops password guessing.
- Sessions expire after 8 hours (one shift); signing out ends the session immediately.

The staff menu



The menu under the staff member's name: role, facility, photo, password, activity log, sign out.



Activity log: who did what, and when.

Scenario: A complaint about privacy

A patient says a staff member gossiped about their diagnosis. The facility manager opens the activity log and sees exactly which staff opened that patient's record and when. Staff know every view is logged, which discourages snooping in the first place.

PART 3

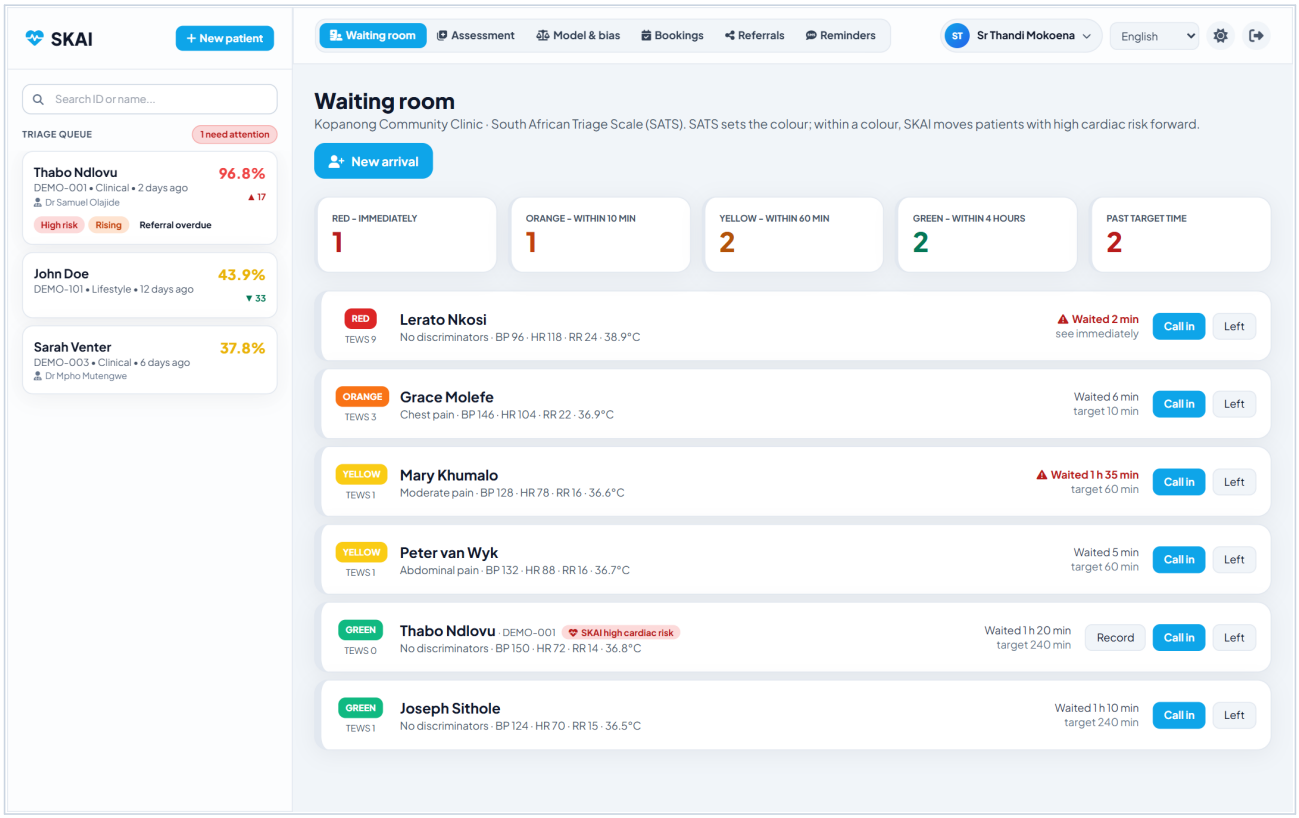
The clinician dashboard

Everything nurses, doctors and managers use

7. The waiting room (SATS triage)

Who uses it: Nurses (doctors can too)

Where in the app: Waiting room tab; nurses land here after signing in

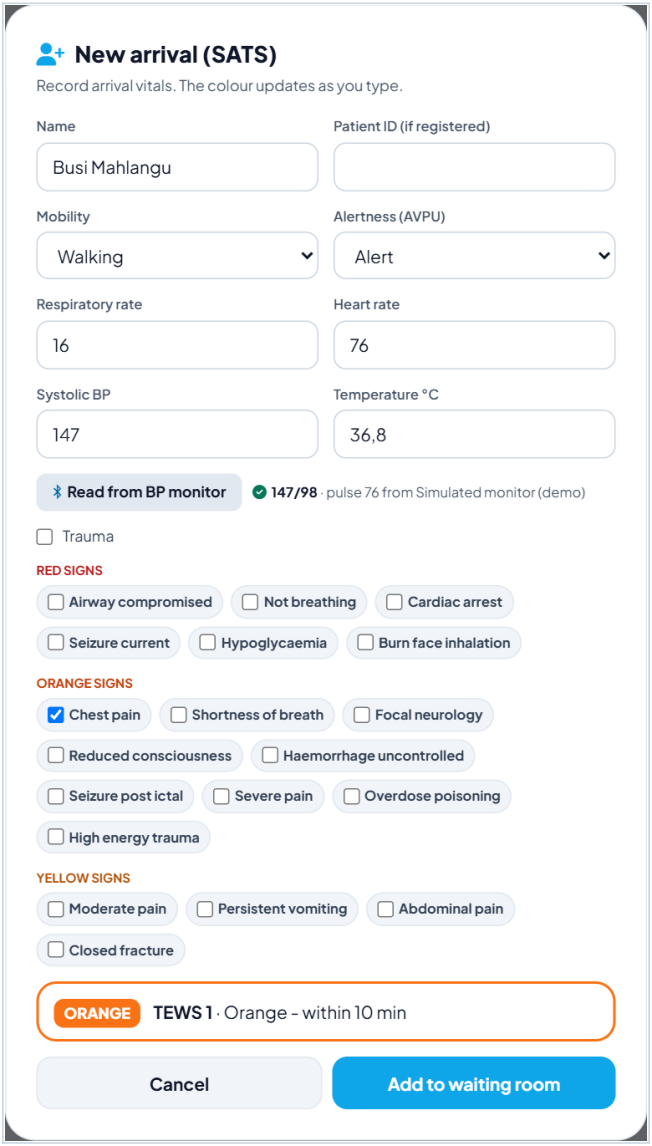


The Kopanong waiting room: counts per colour, patients past their target time, and the queue in safe order.

This screen replaces the paper triage list. At the top are counters: how many people are waiting in each colour, and how many have waited longer than SATS allows (**Past target time**). Below is the queue, in the order patients should be seen:

- **Colour first**: red, then orange, yellow, green. SKAI never changes this.
- **Within the same colour**, a patient whom SKAI rates high or critical for heart risk (red heart badge *SKAI high cardiac risk*) goes first.
- Then whoever arrived first.
- Each row shows the danger signs, vital signs, how long they have waited, and the target time. Overdue rows are red.
- Buttons: **Call in** (patient goes to the consulting room), **Done**, **Left** (left without being seen), and **Record** to open the full patient record. The list refreshes itself every 30 seconds.

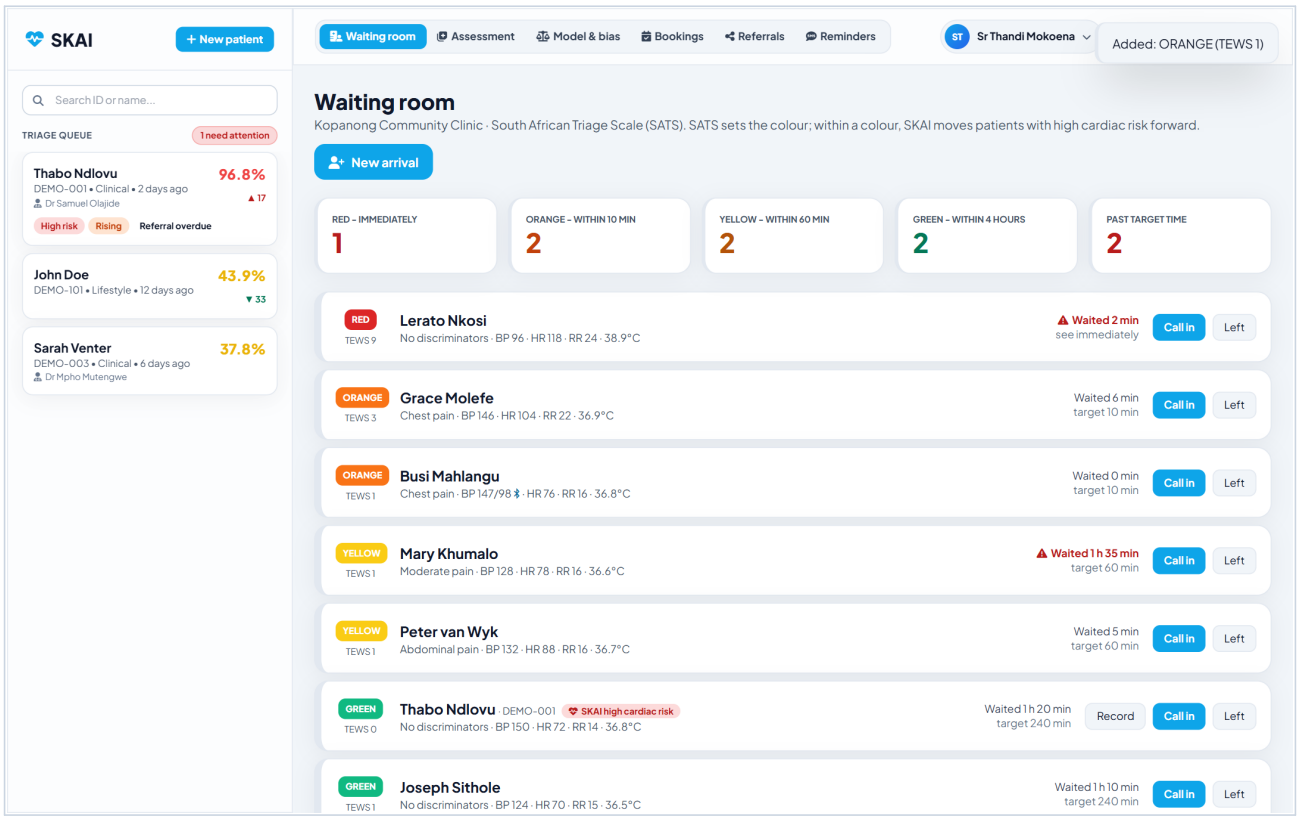
Adding a new arrival



New arrival form: vitals, alertness, mobility and danger signs. The colour updates live as the nurse types.

Scenario: Busi arrives with chest pain

- 08:40. Busi Mahlangu, 52, walks in holding her chest. Sr Thandi clicks **New arrival**.
- She wraps the clinic's Bluetooth blood-pressure cuff on Busi's arm and clicks **Read from BP monitor**. The reading (here 147/98, pulse 76) fills in automatically, marked as device-measured.
- She ticks the danger sign **Chest pain**. Busi's vital signs alone would be green (TEWS 1), but chest pain is an orange discriminator, so the preview box turns **ORANGE: within 10 minutes**.
- She clicks **Add to waiting room**. Busi goes straight above everyone yellow and green, even though they arrived earlier.



After adding Busi: she sits with the orange patients, ahead of all yellow and green patients.

Scenario: Why Thabo goes before Joseph (both green)

Thabo Ndlovu and Joseph Sithole are both green: neither is an emergency today. But Thabo's last SKAI assessment was critical, so he has the SKAI badge and is seen first among the greens. Joseph's sore knee can safely wait; Thabo's silent heart risk should not leave the clinic untreated. Lerato Nkosi (red, feverish and drowsy) is still seen before both of them.

The clinic blood-pressure monitor

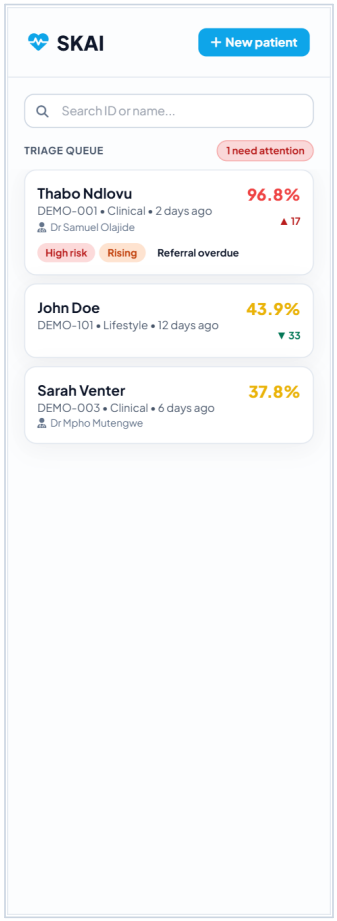
Many clinics already own digital blood-pressure machines. Retyping readings causes mistakes. The **Read from BP monitor** button connects to any cuff that uses the standard Bluetooth blood-pressure profile and copies the reading directly. If the computer has no Bluetooth or no monitor is nearby, SKAI uses a **clearly labelled simulator** so the demo still works.

Limits to be honest about

- The Bluetooth code follows the official standard but has not yet been tested with a physical cuff; the demo uses the simulator.
- SATS here covers adults only. Children have a different SATS chart.
- The waiting-room patients are fictional and reset each morning in demo mode.

8. The patient queue (left sidebar)

Who uses it: Nurses and doctors **Where in the app:** Left side of the Assessment tab



The triage queue: every patient at this facility, most urgent first, with alert flags.

This is the clinic's list of **known patients** (not just today's waiting room), sorted so the most worrying are on top. Each card shows the latest risk, whether it went up or down since last time, who assessed them, and flags:

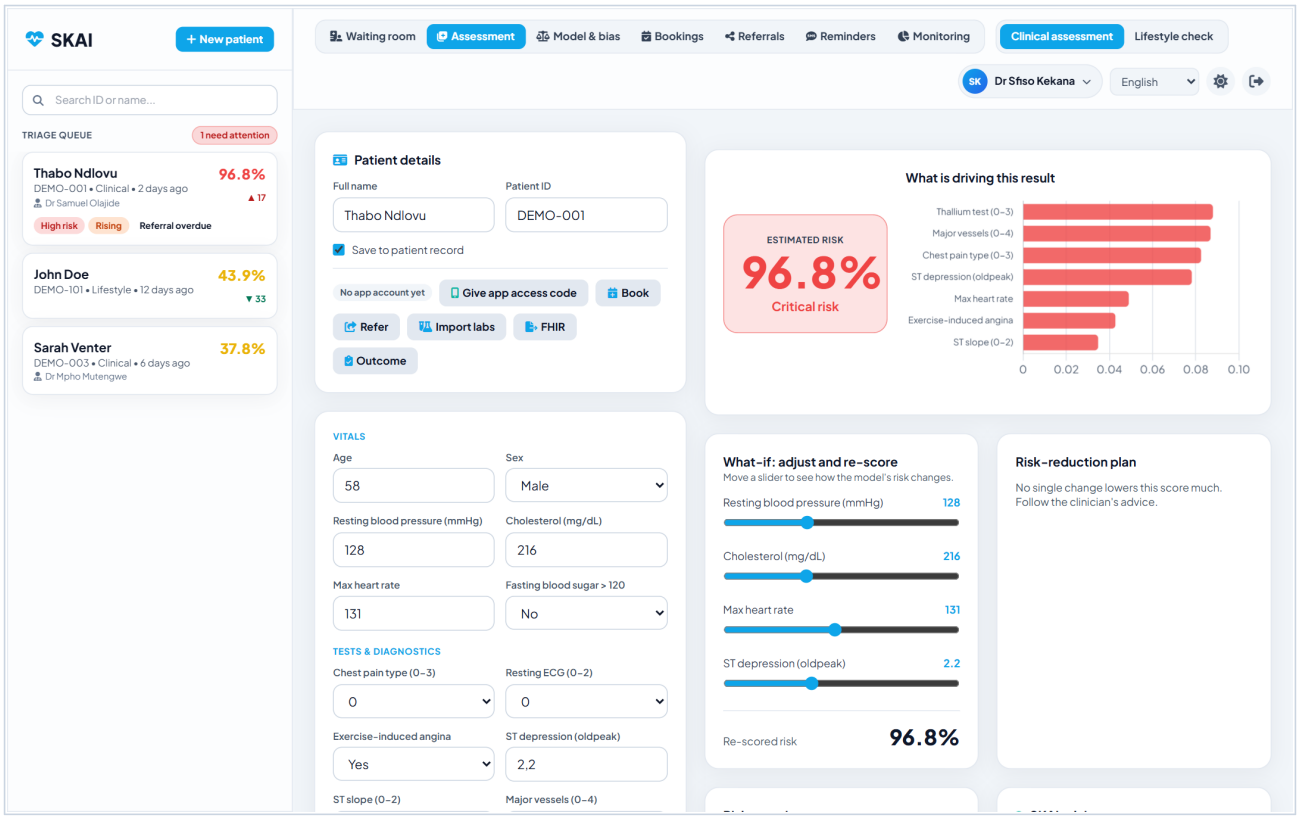
Flag	Meaning	Why it matters
High risk	Latest risk tier is high or critical	Needs action
Rising	Risk went up by 10 points or more since the previous assessment	Something is getting worse
Follow-up due	High-risk patient not reassessed in 30 days	Patients get lost between visits
Verify document	Patient uploaded results that are not digitally signed	A doctor must check them
Referral overdue	A referral was not completed by its due date	The patient may have fallen through the cracks

The search box finds patients by name or ID as you type. **+ New patient** starts an empty record.

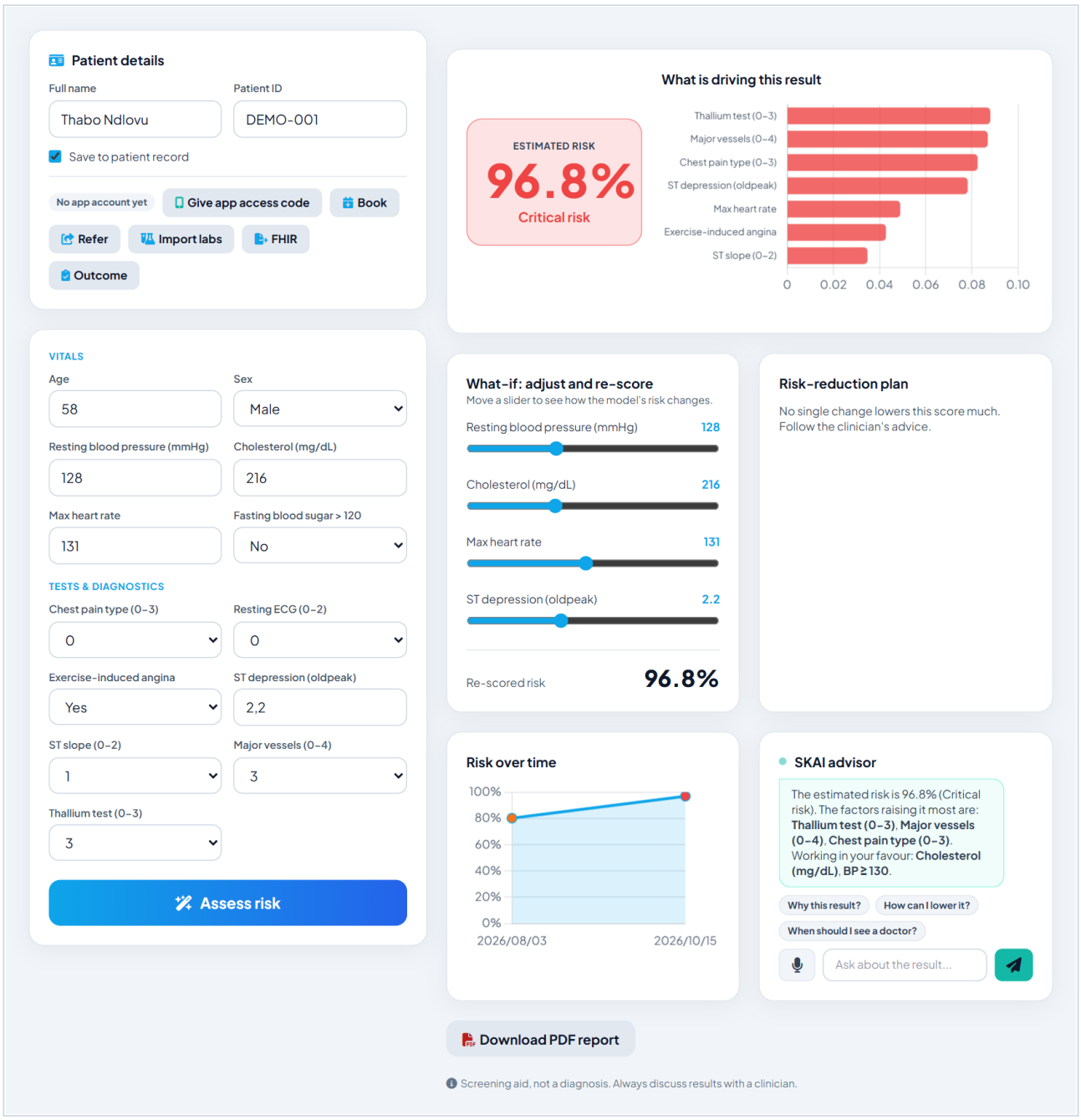
9. Clinical risk assessment

Who uses it: Nurses and doctors

Where in the app: Assessment tab → Clinical assessment



The full workspace with Thabo Ndlovu open: patient details, risk, explanation, what-if sliders and plan.



The assessment panel in detail.

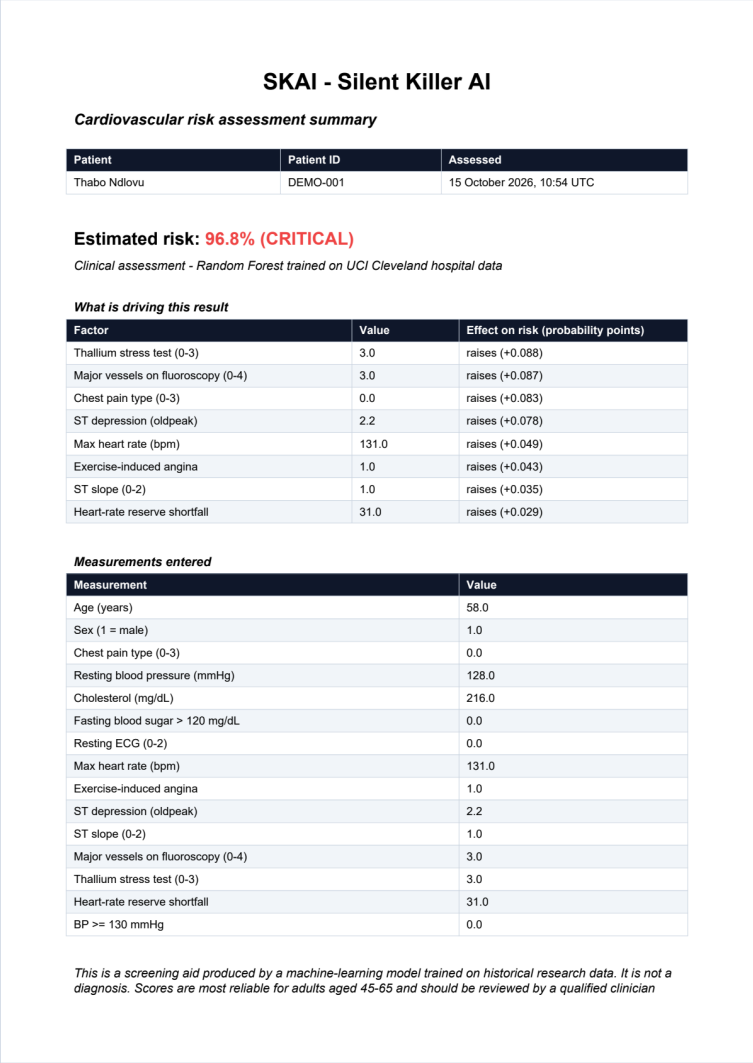
When a patient is opened, SKAI re-runs the model on their saved measurements and shows:

- **Estimated risk** (big number) and the **tier** colour.
- **What is driving this result**: the SHAP bar chart. Red bars push risk up, green bars pull it down. For Thabo, the thallium test, number of major vessels on fluoroscopy and chest-pain type drive his risk most.
- **The form** with all 13 clinical measurements (age, sex, blood pressure, cholesterol, fasting sugar, ECG results, maximum heart rate, exercise-induced chest pain, ST depression, and others). **Assess risk** saves a new assessment.
- **What-if sliders**: move blood pressure, cholesterol, heart rate or ST depression and watch the risk recalculate instantly. Useful for showing a patient how much difference treatment could make. Sliders never change the record.
- **Risk-reduction plan**: the model is re-run with each guideline target (BP 120, cholesterol 200, fasting sugar normal) and lists only the changes that actually lower this patient's risk.
- **Risk over time**: a chart of all past assessments.

- **Download PDF report** (below).

> **Scenario: Explaining the result to Thabo**
>
> Dr Kekana turns the screen towards Thabo. “See these red bars? Your heart tests are the main reason. Your cholesterol is actually helping you a little.” He moves the blood-pressure slider from 128 down to 120: the risk barely moves, because Thabo's risk comes mostly from his heart tests, not his blood pressure. That tells the doctor that a referral for heart investigations matters more than another blood-pressure tablet.

The PDF report



SKAI - Silent Killer AI

Cardiovascular risk assessment summary

Patient	Patient ID	Assessed
Thabo Ndlovu	DEMO-001	15 October 2026, 10:54 UTC

Estimated risk: 96.8% (CRITICAL)

Clinical assessment - Random Forest trained on UCI Cleveland hospital data

What is driving this result

Factor	Value	Effect on risk (probability points)
Thallium stress test (0-3)	3.0	raises (+0.088)
Major vessels on fluoroscopy (0-4)	3.0	raises (+0.087)
Chest pain type (0-3)	0.0	raises (+0.083)
ST depression (oldpeak)	2.2	raises (+0.078)
Max heart rate (bpm)	131.0	raises (+0.049)
Exercise-induced angina	1.0	raises (+0.043)
ST slope (0-2)	1.0	raises (+0.035)
Heart-rate reserve shortfall	31.0	raises (+0.029)

Measurements entered

Measurement	Value
Age (years)	58.0
Sex (1 = male)	1.0
Chest pain type (0-3)	0.0
Resting blood pressure (mmHg)	128.0
Cholesterol (mg/dL)	216.0
Fasting blood sugar > 120 mg/dL	0.0
Resting ECG (0-2)	0.0
Max heart rate (bpm)	131.0
Exercise-induced angina	1.0
ST depression (oldpeak)	2.2
ST slope (0-2)	1.0
Major vessels on fluoroscopy (0-4)	3.0
Thallium stress test (0-3)	3.0
Heart-rate reserve shortfall	31.0
BP >= 130 mmHg	0.0

This is a screening aid produced by a machine-learning model trained on historical research data. It is not a diagnosis. Scores are most reliable for adults aged 45-65 and should be reviewed by a qualified clinician

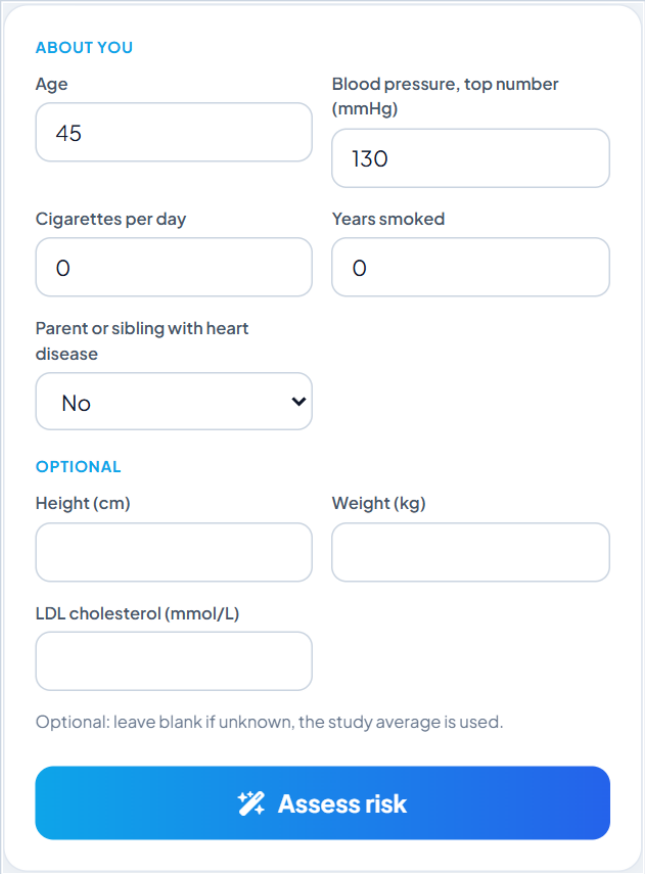
The downloadable assessment report: risk, top drivers and every measurement entered.

Every saved assessment can be downloaded as a one-page PDF to file, print, or give to the patient. Only signed-in staff at the patient's facility, or the patient themself, can download it.

> **Limits to be honest about**
>
> - The clinical model was trained on 303 patients from Cleveland, USA, collected in the 1980s. It has not been tested on South Africans.
> - Several of its 13 inputs (thallium stress test, fluoroscopy, exercise ECG) are **hospital tests**. A primary care clinic usually cannot provide them, so in a clinic the lifestyle check (next section) is the realistic option.
> - Held-out accuracy: catches 78.6% of people with heart disease (recall), 88.0% precision, ROC-AUC 0.92.

10. Lifestyle check (for clinics without heart tests)

Who uses it: Nurses and doctors **Where in the app:** Assessment tab → Lifestyle check (top bar)



The lifestyle check uses only things a clinic can measure or ask.

The second model was trained on the **South African Heart Disease (SAHeart) study**: 462 men from the Western Cape. It needs only age, blood pressure, smoking (cigarettes per day and years), family history, and optionally height, weight and LDL cholesterol. Unknown optional values are filled with the study average. This is the model the patient app uses.

Scenario: A busy Tuesday at Kopanong

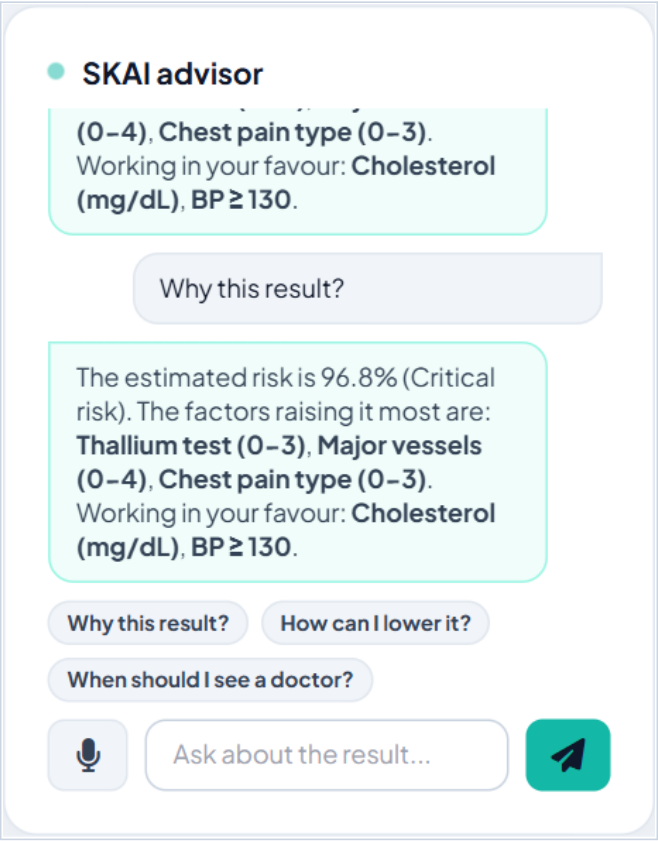
Mr Johan Pretorius, 49, is at the clinic for a flu injection. While he waits, Sr Thandi measures his blood pressure (152) and asks three questions: he smokes 15 a day for 25 years, and his father had a heart attack. The lifestyle check puts him in the high tier. He leaves with an appointment instead of just an injection.

Limits to be honest about

SAHeart included only men, and about 1 in 3 had heart disease, far more than in a normal clinic. Its percentages are therefore too high and it may work less well for women. Held-out results: recall 87.5%, precision 50.9%, ROC-AUC 0.82.

11. The SKAI advisor

Who uses it: Staff and patients	**Where in the app:** Bottom of the Assessment tab; inside the patient app



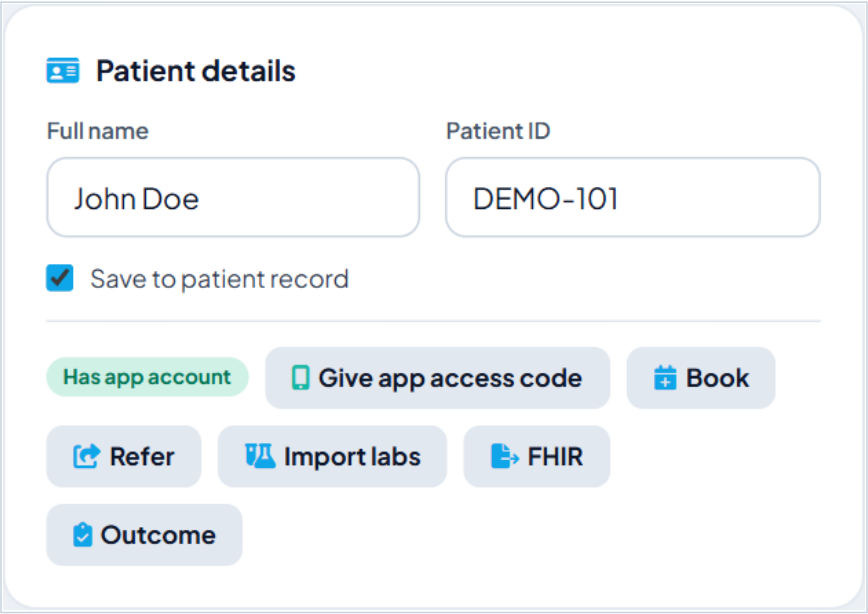
The advisor answers "Why this result?", "How can I lower it?" and "When should I see a doctor?"

The advisor is an assistant that explains the current result in plain language, using the real model output for this patient. Quick buttons cover the three most common questions; a microphone button allows voice questions in browsers that support it. It works **offline**: it does not send patient data to any outside AI service.

> **Limits to be honest about**
>
> It answers from fixed rules based on the model output; it is not a chatbot that can discuss any medical topic.

12. Patient record actions

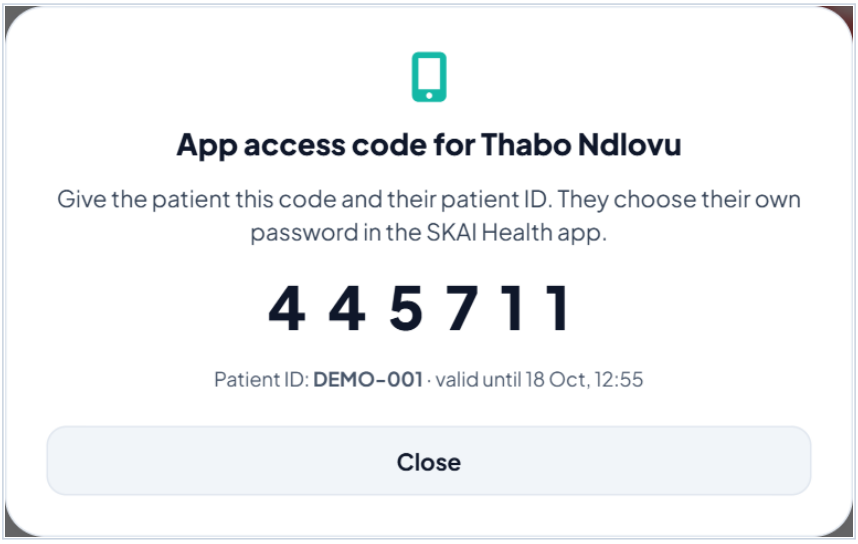
Who uses it: Nurses and doctors (outcome: doctors only)	**Where in the app:** Assessment tab → Patient details card, after opening a patient



Actions available on an open patient record.

Button	What it does
Give app access code	Creates a one-time 6-digit code so a clinic-registered patient can set up SKAI Health (section 19).
Book	Opens the booking form for this patient (section 14).
Refer	Opens the referral form (section 15).
Import labs	Imports lab results in the FHIR standard (below).
FHIR	Downloads the patient's record in the international FHIR format, for another clinic or system.
Outcome	Records what tests later confirmed (doctors only, section 17).

App access codes

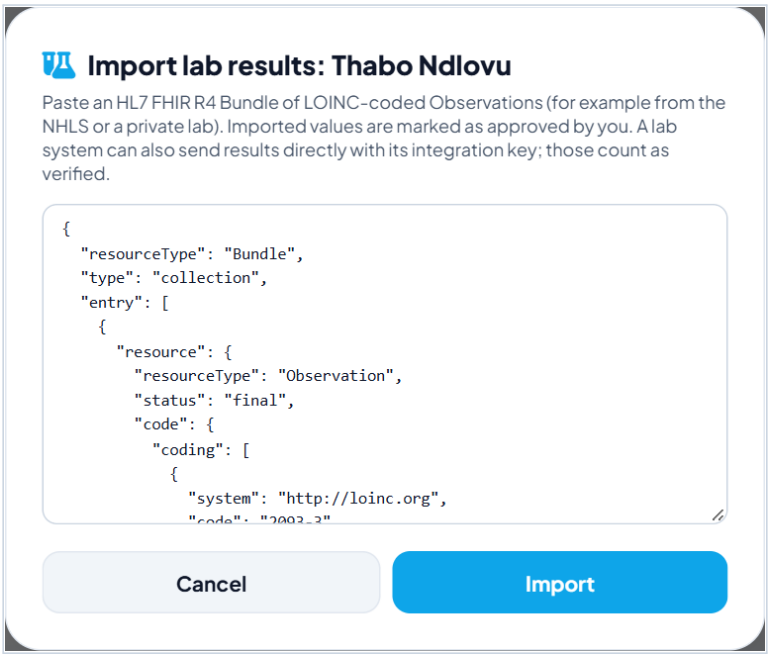


A code for Thabo. It works once, for 24 hours, with 5 tries.

Scenario: Thabo gets the app

Thabo has no email address and has never used an app like this. Before he leaves, Sr Thandi clicks **Give app access code** and writes the code and his patient ID on his clinic card. At home his daughter helps him enter the code and choose his own password. The clinic never knows his password.

Lab results by FHIR



Importing lab results as an HL7 FHIR R4 Bundle (pre-filled with an example).

FHIR (Fast Healthcare Interoperability Resources) is the international standard for exchanging health data, used by labs, hospitals and national systems. Each result is coded with an international **LOINC** code, so "2085-9" always means HDL cholesterol in any system.

- A **lab system** can send results straight to SKAI using its secret integration key. These count as **verified**.
- A **staff member** can paste or import a result file. These count as **approved by that staff member** and are logged.
- Lab values are **added** to the patient's clinic measurements rather than replacing them: blood pressure from the clinic, newer cholesterol from the lab.
- **FHIR export** gives any other system the patient, their measurements and SKAI's risk assessments in the same standard.

Scenario: NHLS results arrive

Thabo's blood was taken last week and sent to the NHLS lab. Instead of a nurse retyping a printout, the lab's system sends a FHIR message: total cholesterol 5.8, HDL 1.1, LDL 3.9, glucose 6.1 mmol/L. SKAI files them under Thabo, and the Framingham score can now use his real cholesterol instead of his weight.

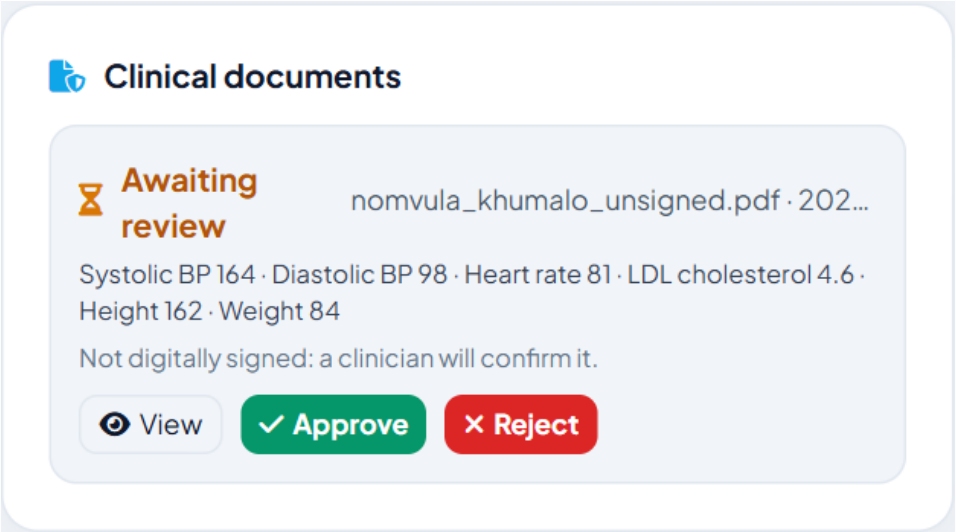
Limits to be honest about

There is no live connection to the NHLS or a private lab yet. That needs an agreement with the lab and access to its systems. The demo uses a demo integration key.

13. Checking patient documents

Who uses it: Doctors approve or reject; all staff can view

Where in the app: Assessment tab → Clinical documents card



Nomvula's clinic printout is not digitally signed, so a doctor must check it.

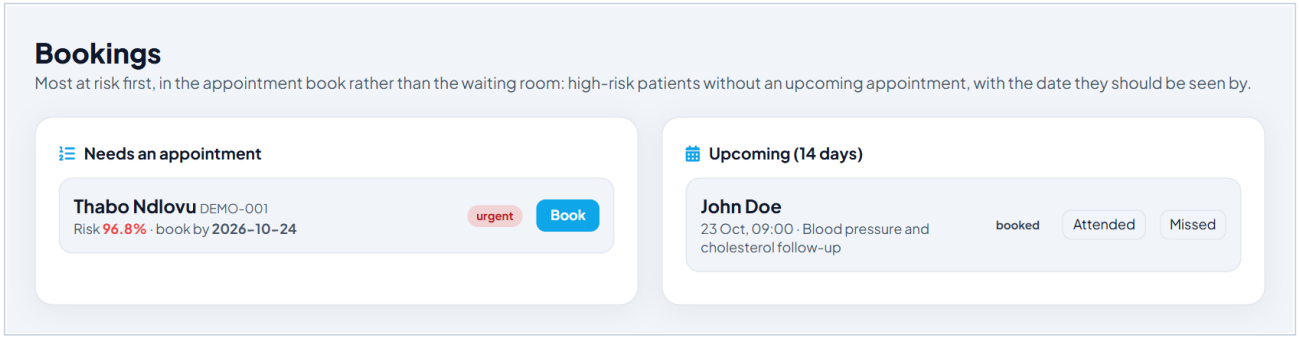
Patients upload their results as PDFs (section 25). If the PDF carries a digital signature from a trusted clinic or lab, SKAI accepts it automatically. If not, SKAI still reads the numbers but marks the document **Awaiting review**; the patient gets a provisional result and the doctor sees a **Verify document** flag.

Scenario: Nomvula's printout

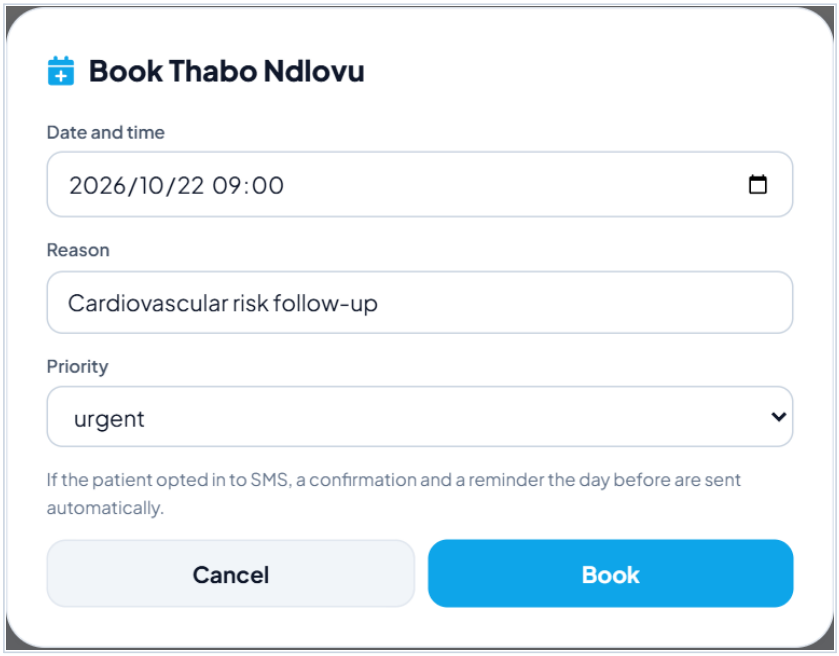
Nomvula uploaded a printout from her clinic visit: BP 164/98. It has no digital signature. Dr Kamogelo clicks **View**, compares the PDF with the clinic file, sees it matches, and clicks **Approve**. Nomvula's result is no longer provisional. Had the numbers been edited, he would click **Reject**.

14. Bookings: most at risk first

Who uses it: Nurses, doctors, facility managers **Where in the app:** Bookings tab



Left: high-risk patients without an appointment and the date to book them by. Right: the next 14 days.



Booking Thabo. If he opted in to SMS, a confirmation and a reminder the day before are sent.

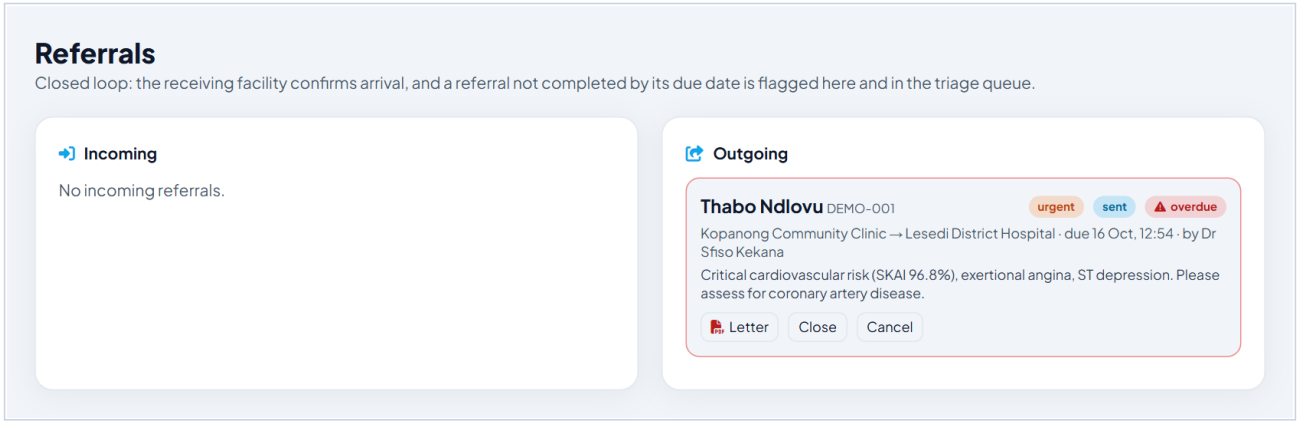
This is where SKAI's original idea, *prioritise the people at most risk*, becomes practical. Moving a high-risk patient ahead in today's waiting room helps a little; giving them a **booked appointment soon** helps much more. The list shows every high or critical patient who has no upcoming appointment, with a **book by** date: within 7 days for critical, 14 days for high. Booked patients drop off the list. Appointments are marked **Attended** or **Missed**.

Scenario: Friday afternoon admin

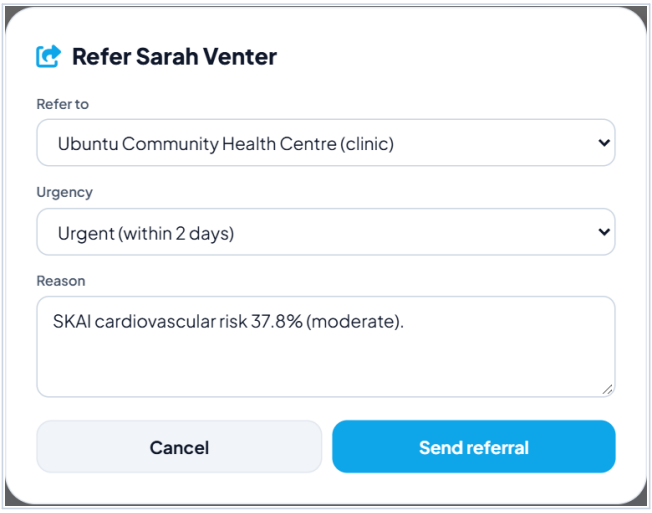
Every Friday, the facility manager at Kopanong opens Bookings. Three high-risk patients have no appointment. She books each into next week's chronic-care slots, and SKAI texts each of them. On Monday the missed appointments show up, and those patients are called.

15. Referrals: closing the loop

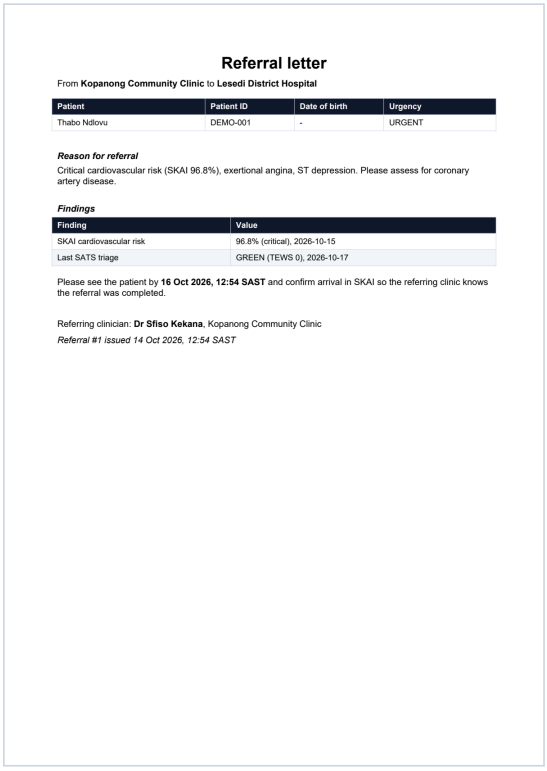
Who uses it: Nurses and doctors at both facilities **Where in the app:** Referrals tab; Refer button on a patient



Outgoing referral from Kopanong to Lesedi District Hospital, marked overdue.



Creating a referral: destination, urgency and reason.



The referral letter PDF, generated automatically.

In South Africa, a common failure is the **open loop**: a clinic refers a patient to hospital, gives them a letter, and never finds out whether they went. SKAI closes the loop:

Sent
Clinic refers; letter printed; SMS to patient

→

Arrived
Hospital confirms the patient came

→

Seen
Hospital has seen them

→

Closed
Clinic closes the loop

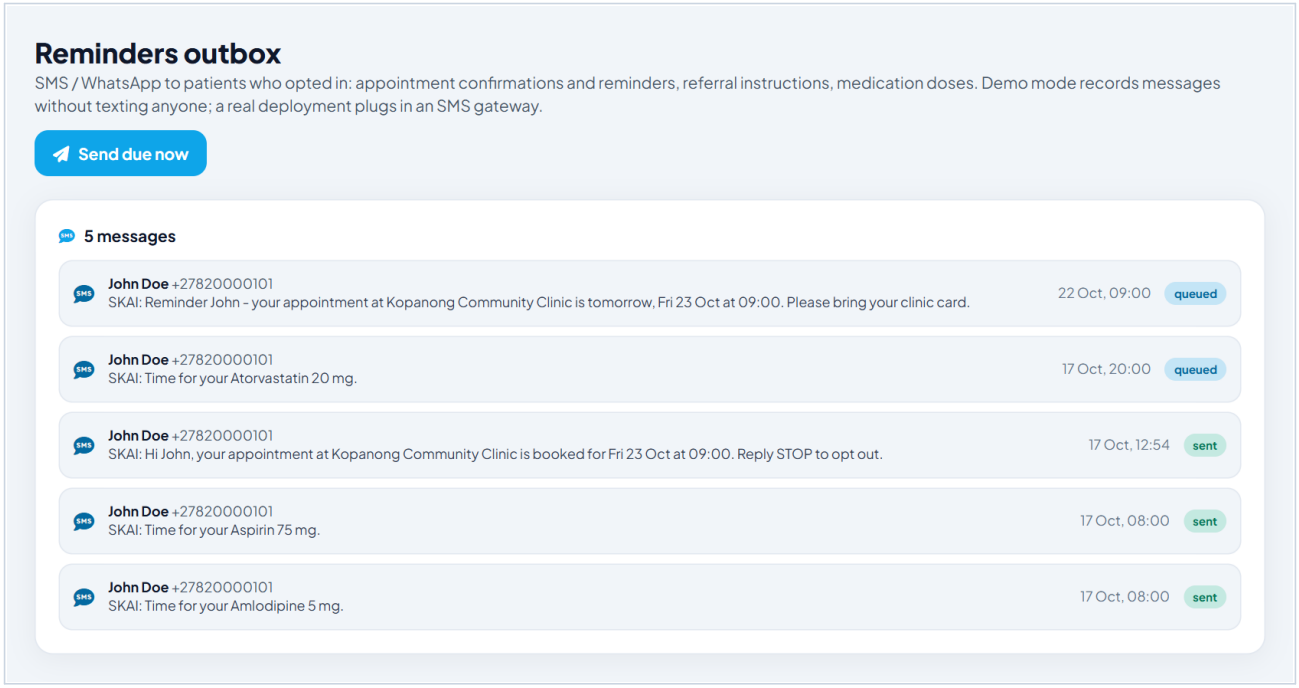
Urgency	Must be completed within
Emergency	6 hours
Urgent	2 days
Routine	14 days

If the hospital has not confirmed arrival by the due date, the referral is marked **overdue** in red, both here and as a flag on the patient in the clinic's queue. Only the receiving facility can confirm arrival; only while the referral is open can the hospital see the patient's record.

> **Scenario: The overdue referral**
>
> Dr Kekana referred Thabo to Lesedi as urgent three days ago. Lesedi never confirmed his arrival. Monday morning, the clinic queue shows **Referral overdue** on Thabo's card. The nurse phones him: he did not have taxi money. The clinic arranges transport. Without SKAI, nobody would have noticed until he had a heart attack.

16. Reminders outbox (SMS and WhatsApp)

Who uses it: All staff can view **Where in the app:** Reminders tab



Every message SKAI has queued or sent: appointment confirmations, reminders and medication doses.

- Messages go only to patients who **opted in** themselves (in the patient app) and gave a valid South African cellphone number.
- Types: appointment booked, reminder the day before, referral instructions, medication dose reminders.
- SKAI checks for due messages every minute; **Send due now** sends them immediately.
- Every message offers "Reply STOP to opt out". Times are in South African time (SAST).

Scenario: John's reminder

John Doe's appointment is on Friday at 09:00. On Thursday at 09:00 his phone buzzes: "SKAI: Reminder John, your appointment at Kopanong Community Clinic is tomorrow, Fri 23 Oct at 09:00. Please bring your clinic card." Missed appointments are one of the biggest reasons chronic patients lose control of their blood pressure.

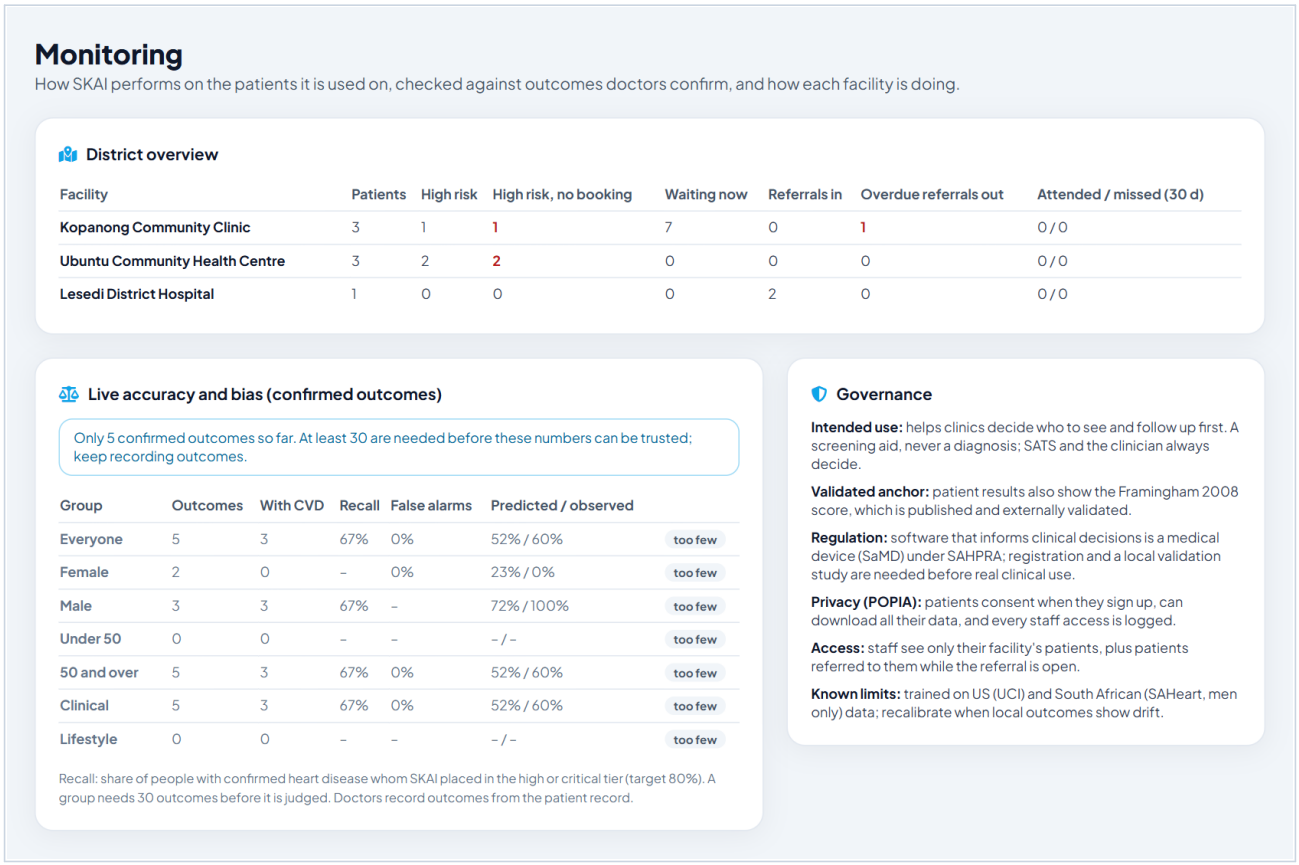
Limits to be honest about

In the demo, messages are recorded but not actually sent. A real deployment needs an account with an SMS or WhatsApp provider (paid per message); the app is built so one can be plugged in.

17. Monitoring and governance

Who uses it: Doctors, facility managers, district manager

Where in the app: Monitoring tab



The district manager's view: facility overview, live accuracy and bias, and the governance summary.

Facility and district overview

One line per facility: number of patients, how many are high risk, **how many high-risk patients have no booking**, people waiting now, referrals coming in, **overdue referrals**, and attended versus missed appointments in the last 30 days. A district manager sees all facilities; a doctor or facility manager sees their own.

> **Scenario: District meeting**
>
> At the monthly district meeting, Ms Dube shows that Ubuntu CHC has two high-risk patients with no booking while Kopanong has one overdue referral. Instead of guessing, the district can see where patients are slipping through.

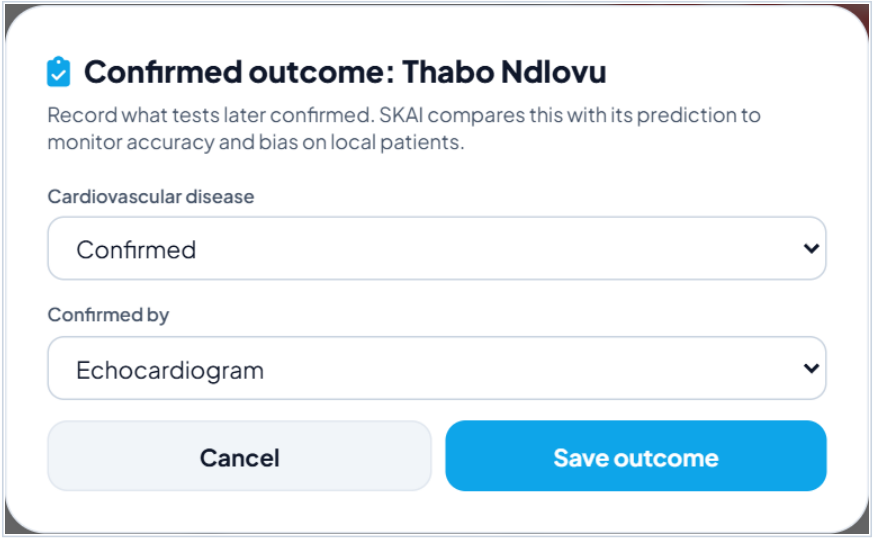
Live accuracy and bias

A model that was accurate on old American data might not be accurate in Tshwane. So when a doctor later confirms whether a patient really has heart disease (echocardiogram, angiogram, ECG, stress test, diagnosis or admission), they press **Outcome** on the patient (section 12). SKAI then compares its prediction with reality:

Measure	Plain meaning
Recall	Of the people who really had heart disease, how many SKAI flagged as high or critical. Target: 80%.

Measure	Plain meaning
False alarms	Of the people without heart disease, how many SKAI wrongly flagged.
Predicted / observed	Average predicted risk versus how many actually had disease. Far apart means the model needs recalibrating.
By group	The same numbers for women versus men, under 50 versus 50 and over, and each model. Big gaps mean possible bias.

With fewer than 30 outcomes in a group, the numbers are shown but marked **too few** to judge, because small numbers can be misleading. The demo has only 6 outcomes, so SKAI itself says it is too early to trust them.



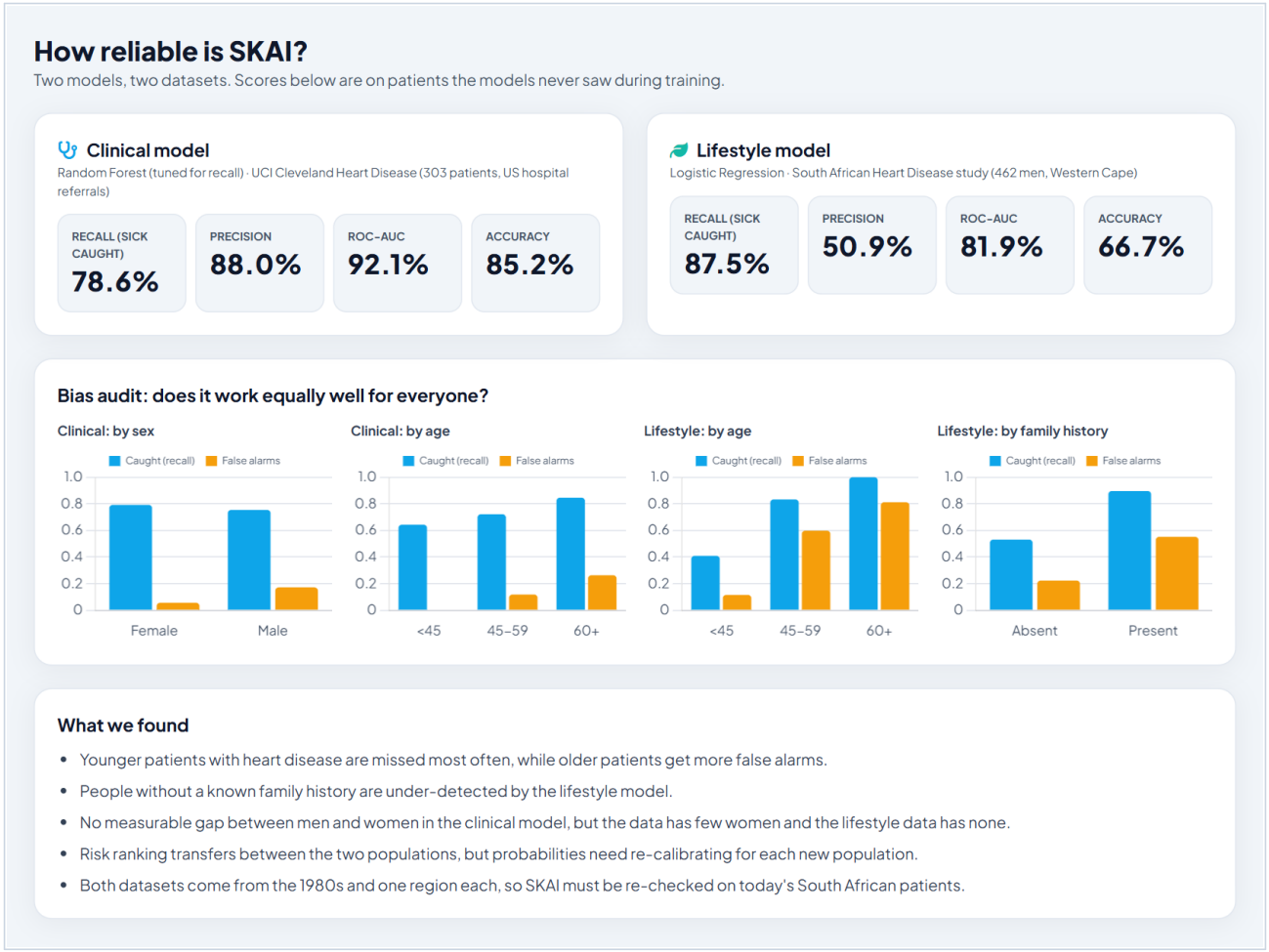
Recording a confirmed outcome for Thabo.

Governance

The governance panel states the rules SKAI runs by: intended use (prioritising follow-up, never diagnosing), the validated Framingham anchor, the need for SAHPRA registration, POPIA privacy protections, facility-based access, and the known limits of the training data.

18. Model & bias page

Who uses it: Everyone (public page) **Where in the app:** Model & bias tab, or the link on the home screen



How reliable SKAI is, and whether it works equally well for everyone, on patients the models never saw in training.

This page answers the question our lecturer asked last time: *what is your bias?* It shows each model's accuracy on held-out patients, then charts how often each group's heart disease is caught (recall) and how often they get false alarms, by sex, by age and by family history. The findings are written underneath in plain words, including:

- Younger patients with heart disease are missed more often, while older patients get more false alarms.
- People without a known family history are under-detected by the lifestyle model.
- No measurable gap between men and women in the clinical model, but the data has few women and the lifestyle data has none.
- Both datasets come from the 1980s and one region each, so SKAI must be re-checked on today's South African patients.

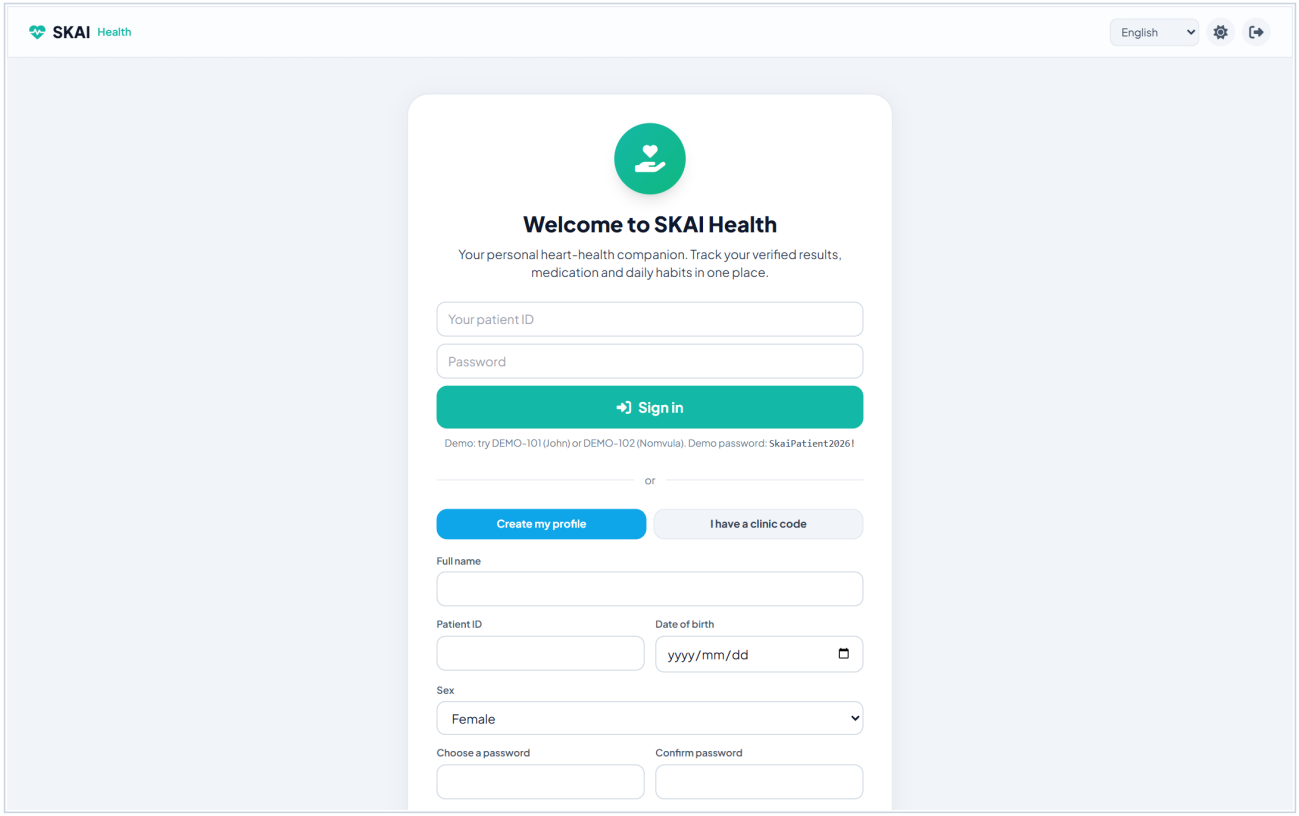
PART 4

SKAI Health: the patient app

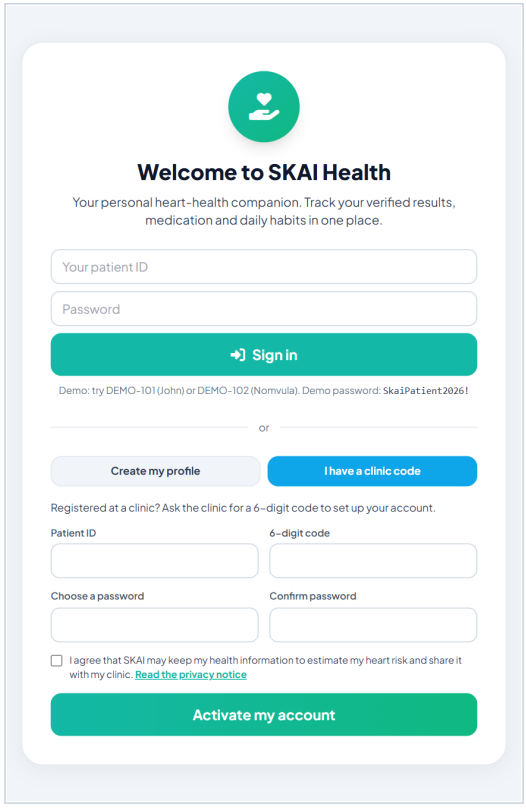
What patients see and do

19. Signing up and privacy consent

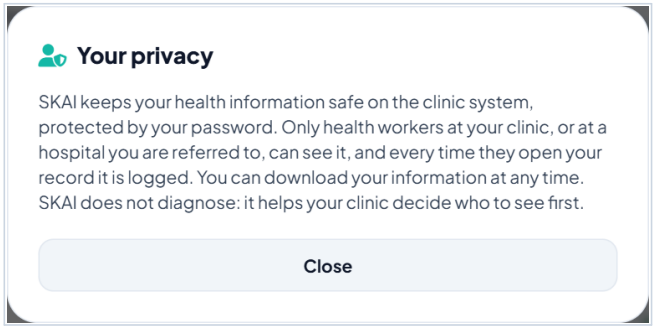
Who uses it: Patients **Where in the app:** Home → Patient self-check



SKAI Health sign-in, and the two ways to get an account.



"I have a clinic code": patient ID, the 6-digit code, a new password, and consent.

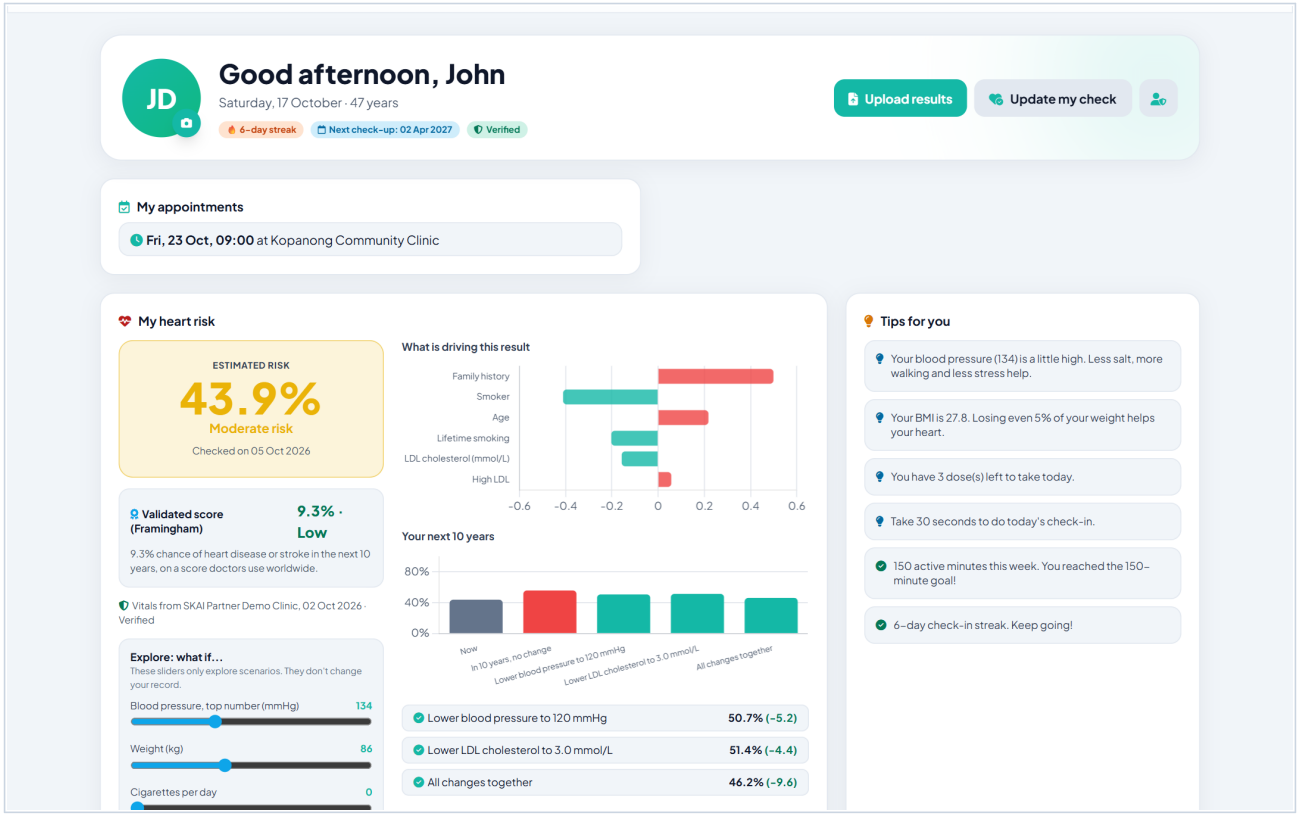


The privacy notice, in the patient's language.

- **Create my profile**: a new patient chooses a patient ID, enters name, date of birth and sex, and a password.
- **I have a clinic code**: a patient the clinic already registered enters the code from the clinic (section 12).
- Either way, the patient must tick **I agree that SKAI may keep my health information…**. Health data is "special personal information" under South Africa's **POPIA** law, so explicit consent is required. SKAI records which version of the notice they agreed to, and when.
- Patients see only their own record. Five wrong passwords lock the account for a while.

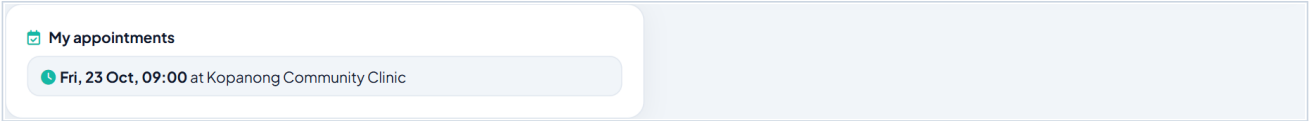
20. The patient dashboard

Who uses it: Patients **Where in the app:** After signing in



John Doe's dashboard: greeting, badges, buttons, appointments, heart risk and tips.

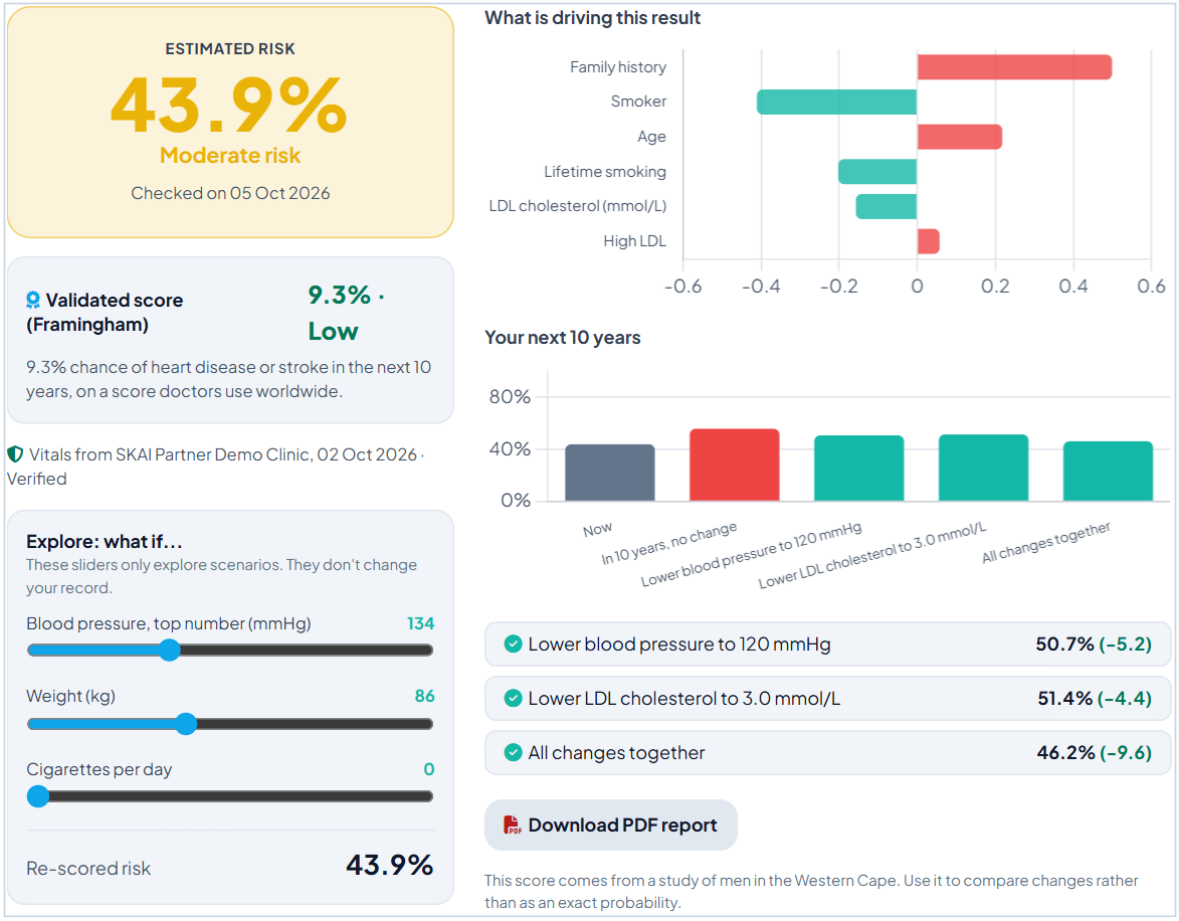
- **Profile photo** (camera button) and a greeting by time of day, with today's date and age.
- **Badges**: check-in streak, next check-up date (from the risk tier), and whether the vitals are verified.
- **Upload results** and **Update my check** buttons, and the shield button for **Privacy and reminders**.
- **My appointments / You have been referred**: shown only when there is one (below).



Upcoming appointments from the clinic's booking (a referral shows here in red with where and by when).

21. My heart risk

Who uses it: Patients | **Where in the app:** Dashboard → My heart risk



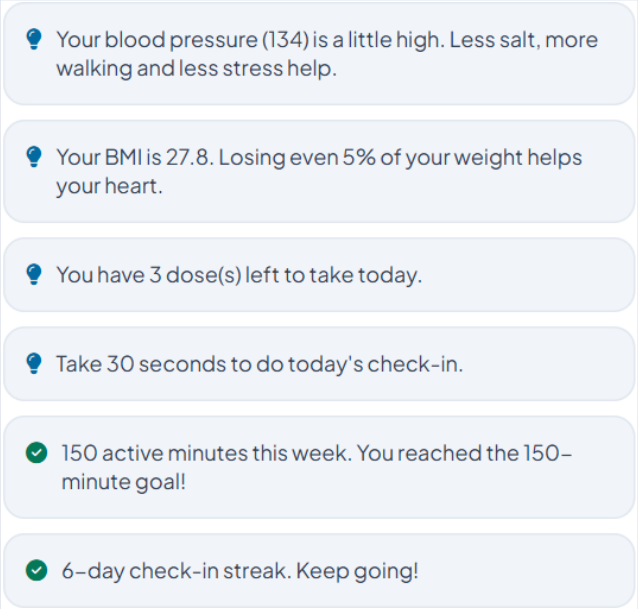
John's risk card: SKAI estimate, the validated Framingham score, explore sliders, drivers and 10-year outlook.

- **Estimated risk** from the lifestyle model, the tier, and the date checked.
- **Validated score (Framingham)**: the published 10-year chance (here 9.3%, low). See section 4 for why the two differ.
- Where the vitals came from (for example *SKAI Partner Demo Clinic, verified*).
- **Explore: what if…** sliders for blood pressure, weight and cigarettes. They never change the record.
- **What is driving this result** and **Your next 10 years**: risk now, if nothing changes, and after each healthy change.
- A checklist of changes with how much each lowers the risk, and a PDF report.

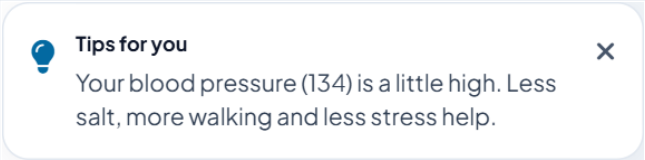
Scenario: John explores

John moves the blood-pressure slider from 134 to 120 and watches his risk fall. Then he moves the weight slider down 8 kg. Seeing his own numbers move is more motivating than a pamphlet.

22. Tips and pop-ups



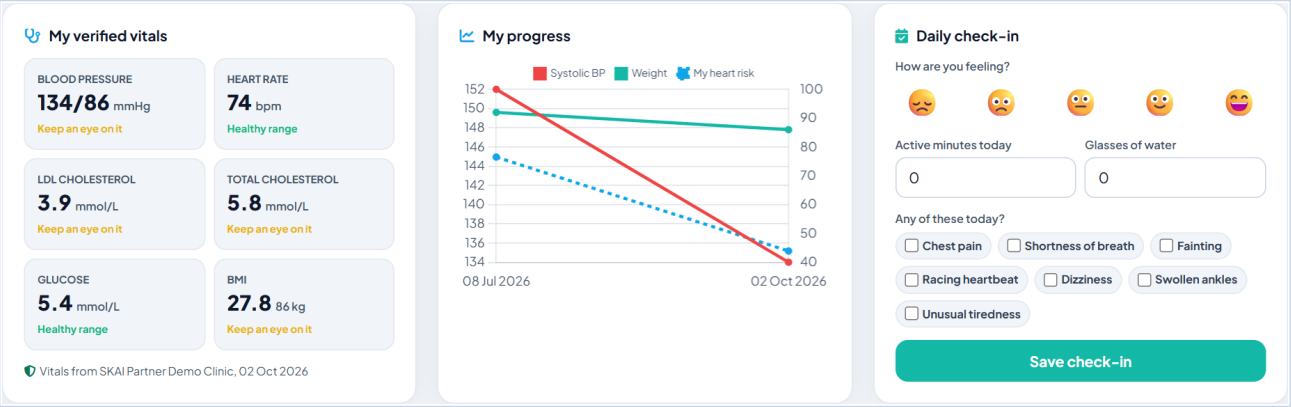
Personal tips based on John's own results and habits.



A tip pops up when he opens the app.

Tips are generated from the patient's own data: blood pressure, BMI, cholesterol, sugar, smoking, medication doses still due today, adherence, activity minutes and check-in streak. Good news is shown too (green), not only warnings. If the patient reports danger symptoms in a check-in, such as chest pain or fainting, an **urgent** screen tells them to call **10177 or 112** immediately.

23. Verified vitals, progress and daily check-in



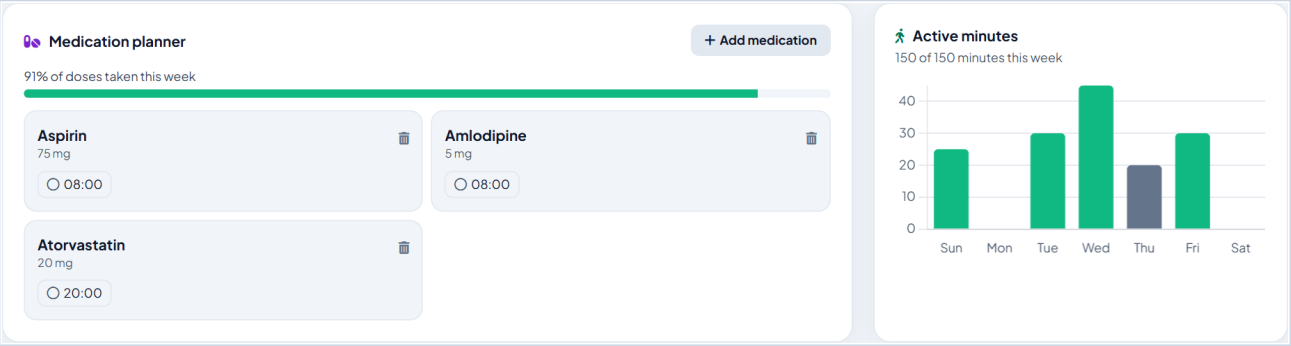
Left: the latest trusted measurements with healthy-range labels. Middle: progress over time. Right: today's check-in.

- **My verified vitals**: blood pressure, heart rate, cholesterol, glucose and BMI, each labelled (for example *Keep an eye on it*).
- **My progress**: blood pressure, weight and risk over time, from each uploaded document.
- **Daily check-in**: mood, active minutes, glasses of water, and symptoms (chest pain, shortness of breath, fainting, racing heart, dizziness, swollen ankles, unusual tiredness). Doing it daily builds a streak.

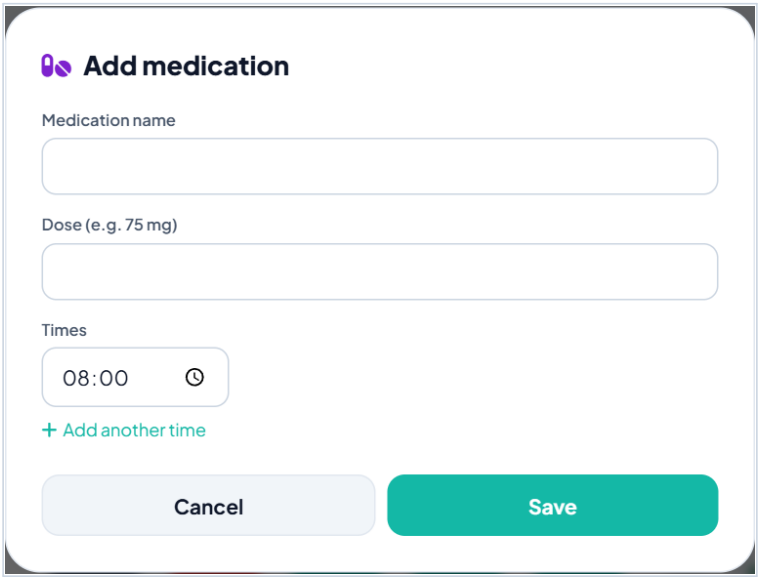
Scenario: A worrying check-in

One evening Nomvula ticks **chest pain** and **shortness of breath** in her check-in. The app immediately shows the heart-attack warning signs and the emergency numbers instead of a friendly tip.

24. Medication planner and active minutes



Medication planner with today's doses to tick off, and active minutes against the WHO 150-minute weekly goal.

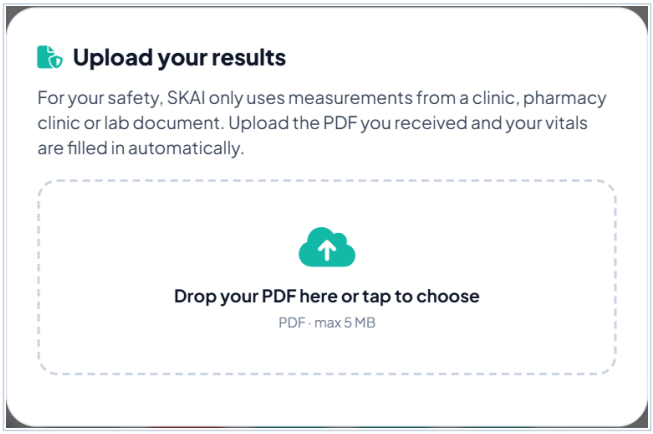


Adding a medication with one or more daily times.

Patients add their medicines and times. Each day they tick doses as taken; the planner shows 7-day adherence. If the patient opted in to SMS, dose reminders are sent at those times. The activity chart compares the week's active minutes with the World Health Organization's goal of 150 minutes a week.

> **Scenario: Forgetting pills**
>
> John's adherence drops to 60% one week. A tip appears: taking blood-pressure medicine every day matters even when you feel fine. His 20:00 Atorvastatin SMS reminder helps him get back on track.

25. Uploading results: how SKAI decides what to trust



Upload screen: the patient uploads the PDF they got from the clinic, pharmacy clinic or lab.



John's documents with their trust level.

Status	When	Used for the risk score?
Verified	The PDF carries a digital signature from a trusted issuer, and it matches. Editing the visible numbers changes nothing.	Yes
Clinician-approved	No signature, but a doctor checked and approved it (section 13), or staff imported it from a lab.	Yes
Awaiting review	No signature; SKAI could read the numbers. A doctor is asked to check.	Provisionally
Rejected	Tampered signature, unknown issuer, future date, someone else's document, or nothing readable. Kept for the record.	No

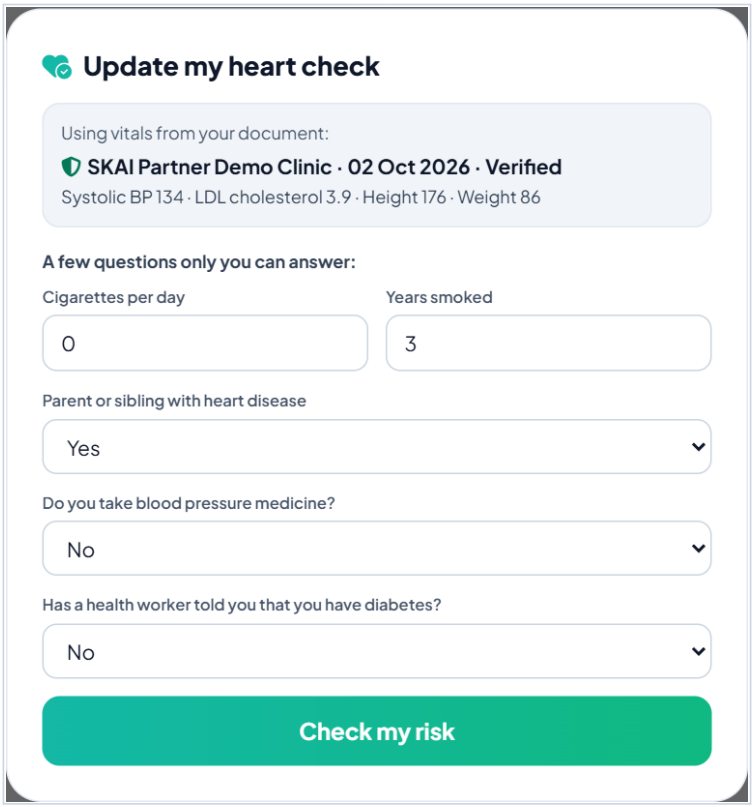
Scenario: Someone tries to cheat

A patient wants a lower risk score for an insurance form, so they edit the blood pressure in their signed clinic PDF from 158 to 118. SKAI checks the signature, sees the document was changed after signing, and rejects it. The signed values cannot be faked.

Limits to be honest about

Photos and scanned paper need text recognition (OCR), which is not built yet. The signing clinic in the demo is fictional; real clinics would need to sign their documents (or send results by FHIR).

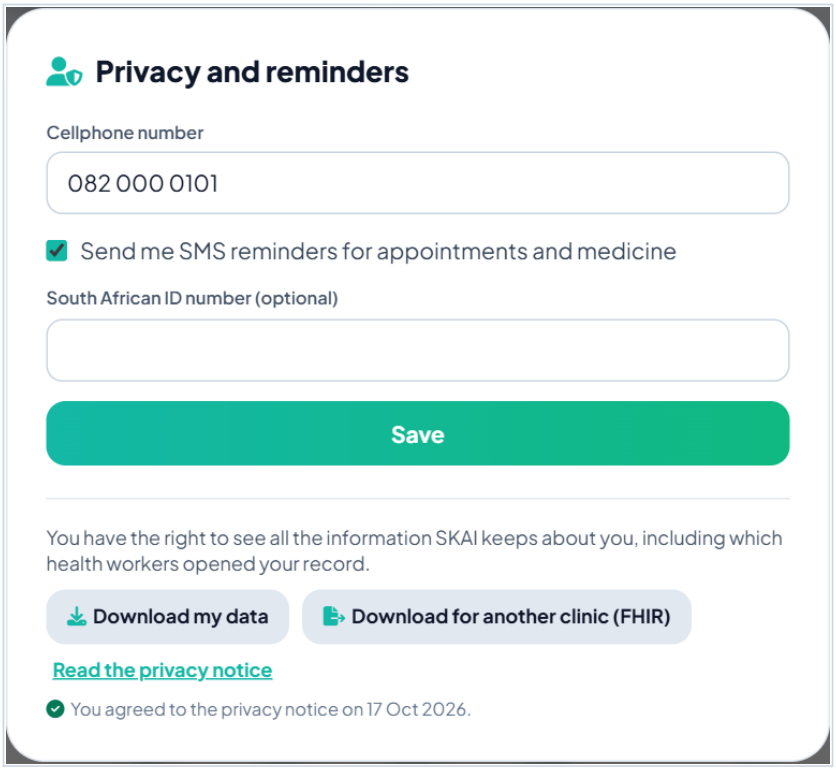
26. Update my check



The check uses vitals from the trusted document and asks only what no document records.

The patient never types blood pressure or cholesterol here; those come from their latest trusted document. They answer only five questions: cigarettes per day, years smoked, family history of heart disease, whether they take blood-pressure medicine, and whether a health worker told them they have diabetes. The last two are needed by the Framingham score.

27. Privacy and reminders settings

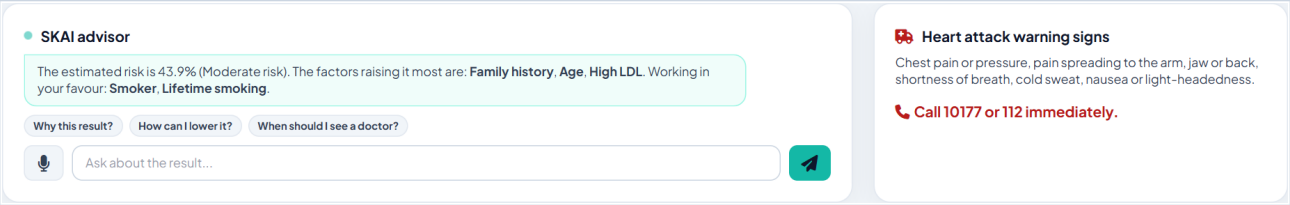


Phone number, SMS opt-in, optional SA ID number, and download buttons.

- **Cellphone number** (checked to be a valid South African mobile number) and **SMS reminders** on or off. Off by default.
- **South African ID number** (optional): checked with the ID's built-in check digit, and the date of birth and sex inside the number must match the profile, which catches typing mistakes and mix-ups between patients.
- **Download my data**: everything SKAI holds about the patient, including **which staff opened their record** (their right of access under POPIA).
- **Download for another clinic (FHIR)**: their record in the international standard, so it can move with them.
- When they agreed to the privacy notice.

> **Scenario: Moving to Johannesburg**
>
> Nomvula moves for work. She downloads her FHIR record and her new clinic can import her results and history, instead of starting again from nothing.

28. Advisor and warning signs in the patient app



The same offline advisor, explaining John's result to him; and the heart-attack warning signs with emergency numbers.

PART 5

Honest limitations

What SKAI cannot do yet, and what in the demo is simulated

29. Limitations

Data and models

- **Not tested on South Africans in practice.** The clinical model learned from 303 patients in Cleveland, USA (1980s); the lifestyle model from 462 men in the Western Cape (1980s).
- **The lifestyle model over-estimates**, because its training study deliberately included many men with heart disease.
- **Women are under-represented**: few in the clinical data, none in SAHeart.
- **The clinical model needs hospital tests** (thallium scan, fluoroscopy, exercise ECG) that clinics do not have.
- **Framingham was built on Americans** and has not been validated in South African patients.
- Risk models estimate probability; they will always miss some sick people and flag some healthy ones.

Clinical and regulatory

- Software that influences clinical decisions is a **medical device**. SKAI would need SAHPRA registration before real use.
- It needs ethics approval and a proper study before it touches real patient care.
- SATS rules are for adults; children are not covered.

Technical

- It needs an internet or network connection; load shedding and poor connectivity would need an offline mode.
- The database (SQLite, one file) is fine for a demo, not for a district. A production version needs a proper server database, secure hosting in South Africa, backups and an independent security test.
- It does not yet connect to the national patient registration system or the systems clinics already use, which would mean double data entry.
- Translations need native-speaker review; newer staff screens are English only.
- Uploaded photos and scans cannot be read yet (OCR).

30. What is real and what is simulated in the demo

Real (works as it would in practice)	Simulated or demo-only
Both trained models, their explanations (SHAP), what-if and plans	All patients, staff and facilities are fictional
SATS / TEWS calculation and the waiting-room order	SMS messages are recorded, not actually sent
Framingham 2008 formula (matches the published worked examples)	BP monitor uses a simulator unless a Bluetooth cuff is connected (untested with a real cuff)
Digital-signature checking of uploaded PDFs	The trusted signing clinic is fictional
Sign-in security, roles, facility access rules, audit log	Lab feed uses a demo integration key; no real lab connected
FHIR export/import format, SA ID check, PDF reports and letters	Outcomes are a handful of demo cases (too few to judge)
POPIA consent recording and data download	Demo waiting room resets each morning

95 automated tests check these behaviours whenever the code changes, for example the waiting-room order, that a nurse cannot approve documents, and that tampered PDFs are rejected.

PART 6

From prototype to the health system

A realistic plan of action, starting in Tshwane

31. Could SKAI really be used in South Africa?

Yes, potentially, but not as it is today, and not quickly. The design fits how South African clinics already work (SATS, referral pathways, chronic-care follow-up, POPIA) and targets a real problem: high blood pressure that goes undetected or uncontrolled. What it lacks is evidence on local patients, regulatory approval, and production-grade engineering. Any honest plan starts small, proves safety first, and earns trust step by step.

Why Tshwane first

- A large metro district with both city and peri-urban clinics (for example around Mamelodi, Soshanguve, Atteridgeville) and district and academic hospitals to refer to, so the full referral loop can be tested.
- Two health-sciences universities nearby (the University of Pretoria and Sefako Makgatho Health Sciences University in Ga-Rankuwa) that could lead the research and ethics side.
- Close to the national Department of Health and SAHPRA in Pretoria.

32. Plan of action

The timeline is indicative. Each phase only starts if the previous one shows SKAI is safe and useful.

Phase	What happens	What must be true to move on
0. Now (this project)	Working prototype; bias audit; honest limits documented.	Presentation done; feedback gathered.
1. Partners and approvals about 6 months	Partner with a university and the Tshwane District Health Services. University ethics approval and provincial research approval. POPIA: appoint an information officer, sign data agreements, host data in South Africa. Engineering: proper database, security test, offline mode for load shedding, link to the national patient registration number.	Ethics and district approval granted; security test passed.
2. Silent-mode pilot about 12 to 18 months	2 to 3 Tshwane clinics and their referral hospital. SKAI runs alongside normal care but **does not change anyone's treatment**. Nurses enter data; doctors record outcomes over 12 months. The Monitoring tab measures accuracy, calibration and bias on local patients, plus how much time it costs nurses.	Enough outcomes (hundreds) to judge accuracy by sex and age; staff find it workable.
3. Make it South African	Recalibrate or retrain the models on the pilot data so percentages match Tshwane patients. Independent review of the new models and bias results.	Recalibrated model meets recall and fairness targets on new local data.
4. Live pilot	Switch on the low-risk process features first: bookings, referral tracking, SMS reminders. Then risk-guided prioritisation within SATS. Compare with similar clinics that do not use SKAI.	Better results than comparison clinics, e.g. more referrals completed, fewer missed appointments, more patients with controlled blood pressure, and no safety problems.
5. Register and scale	SAHPRA registration as software as a medical device. Cost-effectiveness study. Agreement with the Gauteng Department of Health and alignment with National Health Insurance plans. Roll out across Tshwane, then Gauteng, then other provinces.	SAHPRA approval; funding; support and training capacity.

What we would measure

- Recall: how many people with heart disease SKAI flags (target 80% or more), equally across women and men, young and old.
- Calibration: predicted risk matches what actually happens.

- Referrals completed on time; appointments attended; time from high-risk flag to treatment.
- Share of high-risk patients with blood pressure under control after 6 and 12 months.
- Nurse time per patient and staff satisfaction (a tool nobody has time to use helps nobody).

Risks and how we would handle them

Risk	How we would handle it
Model is wrong for local patients	Silent mode first; recalibrate before it influences care; ongoing monitoring with alerts
Bias against women or other groups	Monitor by group from day one; stop or adjust if gaps appear
Extra work for overworked nurses	Bluetooth BP cuffs, lab feeds and links to existing systems to avoid retyping; measure time
Load shedding and connectivity	Offline mode that syncs later; works on low-cost devices
Privacy breach	POPIA compliance, encryption, SA hosting, access logs, least-access roles, security testing
Alert fatigue (too many flags)	Keep flags few and actionable; review with nurses

33. For the presentation

Lines to use

- “SKAI doesn't replace triage. SATS decides who is urgent today; SKAI makes sure the people with the most dangerous hidden problem are booked, referred and followed up.”
- “Patients can't type their own vitals. Every number comes from a trusted clinic, lab or device.”
- “Our bias audit shows where SKAI is weaker, and the monitoring tab keeps checking that on real patients.”
- “SKAI isn't ready to treat patients, and it shouldn't be until it is tested on South Africans. The next step would be a silent-mode pilot in a few Tshwane clinics with a university and the district.”

Likely questions

Question	Answer
Why are SKAI's and Framingham's numbers so different?	SKAI's lifestyle model learned from a study with many sick men, so it over-estimates; we use it to rank patients. Framingham is the realistic reference. See section 4.
Can it work with more than one dataset?	Yes: two models on two datasets (UCI Cleveland and South African SAHeart), plus lab data by FHIR and outcomes collected locally for recalibration.
What is your bias?	Fewer women in the data; younger patients missed more; 1980s data from two regions. Shown on the Model & bias page and monitored live.
Is patient data safe?	Hashed passwords, 8-hour sessions, lockouts, facility-based access, audit log, POPIA consent, data download.
Why not just use Framingham?	It was built on Americans, needs cholesterol, and does not do triage, bookings, referrals, reminders or monitoring. SKAI is the system around the score.
What about clinics without internet?	An offline mode is part of phase 1 of the plan.

Demo checklist

- Restart the server on the morning (**python run_app.py**) so the waiting room shows fresh “this morning” patients.
- Nurse view: **nurse.thandi**. Doctor: **sfiso.kekana**. District: **manager.district**. Password **SkaiDemo2026!**

- Patient app: **DEMO-101**, password **SkaiPatient2026!**
- Suggested order: waiting room (add Busi) → Thabo's assessment → refer → referrals (overdue) → bookings → monitoring → patient app (risk, Framingham, settings) → model & bias.

Glossary

Term	Meaning
AVPU	Alertness scale: Alert, responds to Voice, responds to Pain, Unresponsive
BMI	Body mass index: weight compared with height
Calibration	Whether predicted percentages match what really happens
FHIR / LOINC	International standard for exchanging health data / standard codes for lab tests and measurements
Framingham score	A published 10-year heart-disease risk formula from a long-running US study
HDL / LDL	“Good” and “bad” cholesterol
POPIA	Protection of Personal Information Act: South Africa's privacy law
Recall (sensitivity)	Share of people who really have the disease that the model catches
SAHPRA	South African Health Products Regulatory Authority: approves medicines and medical devices, including software
SaMD	Software as a medical device
SATS / TEWS	South African Triage Scale / its Triage Early Warning Score
SHAP	Method that shows how much each measurement pushed a prediction up or down
Silent mode	Running a tool alongside normal care without letting it change decisions, to test it safely